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INFORMATION TECHNOLOGY **9626/12**

Paper 1 Theory **February/March 2019**

1 hour 45 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your centre number, candidate number and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

Calculators must not be used on this paper.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Any businesses described in this paper are entirely fictitious.

DC (JP/CB) 164153/3
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1 Tick the **four** statements, referring to different types of application software, which are true.

	✓
You can use word processing software to create tables of data	
You can only use database software to create reports	
Spreadsheet software can be used to produce charts	
You must always use web authoring software when creating web pages	
Financial modelling can involve the use of spreadsheet software	
Video editing software cannot be used to edit a music soundtrack	
Word processing software can include multimedia features	
You can only create data forms using database software	
Photo editing software is only used to edit photos	
A management information system (MIS) is a form of database management system	

[4]

2 Tick the **four** most accurate statements, referring to the digital divide.

	✓
The digital divide can exist between the educated and uneducated	
The digital divide can exist between those living in rural areas and those living in urban areas	
The digital divide does not exist between people from different economic classes	
Everyone these days owns a smartphone so the divide is closing rapidly	
Minority groups have no difficulty performing transactions on smartphones thereby reducing the divide	
Disabilities of potential internet users constitute another type of digital divide	
Some less developed countries have limited access to digital services	
Most tribal communities have universal mobile phone coverage in their community	
The digital divide only refers to those who can and cannot access the internet	
The price of connection to the internet tends not to vary between countries	

[4]

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3 A student is researching the topic of her new assignment. She has been provided with a series of hard copy magazine articles. She has been told by her tutor that these are probably more reliable than searching the World Wide Web for information.

Compare and contrast the use of these two types of data sources for her research.

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4 There are many types of systems software including interpreters and device drivers.

(a) In order for a computer to output to a printer it uses a printer driver. This is just one of many device drivers a computer uses.

Explain the purpose of device drivers.

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(b) Describe the function of an interpreter.

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5 A teacher has reminded her students to proof read their assignments before they hand them in.

Define the term 'proof reading'.

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6 Many mobile phone companies, in the past, persuaded customers to move from 3G to 4G.

Compare and contrast 3G and 4G technologies.

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7 Mumbai Stores is a company which has a chain of shops. The company stores the customers' reference number, and the value of the goods they have purchased, in a transaction file. Any payments the customers have made on their account at the end of the month are also stored in the same file.

Parts of the Transaction file and Master file are shown.

The Goods_bought field contains only the value of goods bought using the account's credit facilities.

The Master file shows the customers reference number, the money they owe from the previous month (Old_balance) and the money they now owe (New_balance).

The New_balance has to be calculated by using the Old_balance, Goods_bought and any Payment made. All these values are in rupees.

Transaction file

Customer_ref_no	Goods_bought	Payment
943201	5500	4000
256431	3800	3800
319852	9200	10000
433303	3200	5200
590871	4300	6800
612759	3700	3700
650897	2300	2300
698915	4000	6600
751654	4400	4400
804782	5200	8200

Master file

Customer_ref_no	Old_balance	New_balance
256431	0	
319852	10000	
433303	5000	
590871	4500	
599812	0	
612759	0	
650897	0	
678945	6800	
698915	2600	
712356	5100	
751654	0	
804782	4200	
943201	0	
963584	2400	

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(a) Describe the processes that must happen to the Transaction file before the Master file can be updated.

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(b) Using this data, explain how the Master file is updated using the Transaction file shown. You may assume that the only transaction being carried out is the calculation of the New_balance.

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(c) Each customer has a credit limit of 20000₹. Customers must make a payment of at least 2000₹ each month.

Using examples from the Transaction file, explain why it would be appropriate to use a range check on one field and a limit check on another field.

Please note, the Customer_ref_no is stored as text.

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8 Rishu has been asked by the directors of the company she works for to set up a fully functioning video-conferencing room. It will be used by several workers at the same time to communicate with the directors who will be in a different building.

Describe, including a use of each item, the hardware she will need to purchase.

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9 Describe the differences between health issues and safety (apart from e-safety) issues experienced by employees in an IT-based work environment.

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10 Jasvir owns a gardening equipment hire shop. She has developed a simple database system which contains information about the equipment. Each record contains the following data: Equipment_name, Manufacturer, Equipment_code (a six character code), Hire_price, Number_of_days_hired, Date_hired, Customer_number and Loyalty_customer (Y/N).

Design an appropriate online data entry form for **one** item of equipment, clearly indicating the amount of space required for each field. Use the computer screen shown below for your design. Where appropriate allow the user to enter data by clicking the mouse rather than entering text.

[8]

11 Juanita is the company secretary of a small business in the USA. She has created a spreadsheet to calculate the payroll.
The spreadsheet is shown below.

	A	B	C	D	E	F	G	H	I	J	K	L
1							Roles					
2							L	F	OS	M	D	
3							Labourer	Fitter	Office secretary	Manager	Director	
4							$15	$25	$22	$40	$80	Hourly rate
5		Job codes	Job title	Hours worked	Weekly pay							
6												
7	John Vercoe	L	Labourer	40	$600							
8	Lauren Hall	OS	Office secretary	38	$836							
9	Chloe Rhodes	F	Fitter	39	$975							
10	Owen Williams	OS	Office secretary	42	$924							
11	Lotte Charles	L	Labourer	35	$525							
12	Jeannie Burke	OS	Office secretary	38	$836							
13	Jimmy Lee	F	Fitter	40	$1,000							
14	Dawid Jones	F	Fitter	32	$800							
15	Li Sun Wu	M	Manager	45	$1,800							
16	Charlotte Watson	D	Director	50	$4,000							
17	Joe Norfolk	M	Manager	41	$1,640							
18	Joanne Floyd	F	Fitter	38	$950							
19	Anthony Campbell	F	Fitter	40	$1,000							
20	Linda Stuart	L	Labourer	38	$570							
21	Kate Smith	L	Labourer	37	$555							
22	Sam Smith	OS	Office secretary	40	$880							
23	Duncan Rydell	L	Labourer	36	$540							
24	Tony Thomson	L	Labourer	37	$555							
25	Dougie Ryder	OS	Office secretary	41	$902							
26												

(a) She has created a named range **Roles**

Describe how she created this named range.

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(b) She entered a formula in E7 using the named range and the information in column B to calculate the weekly pay for John Vercoe.

Write down the formula she used. This formula should be easy to replicate.

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..... [6]

You can use the space below for any working you need.

12 A recent development in backing storage of laptop computers has been the increased use of solid state drives (SSDs). However hard disk drives (HDDs) are still used in many laptops.

Evaluate the use of SSDs and HDDs for storing data in laptop computers.

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13 Encryption can be used to protect data on a hard disk and in emails.

Evaluate the need for this encryption.

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INFORMATION TECHNOLOGY **9626/12**

Paper 1 Theory **February/March 2020**

1 hour 45 minutes

You must answer on the question paper.

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INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
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INFORMATION

- The total mark for this paper is 90.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages. Blank pages are indicated.

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[Turn over

1 **(a)** Tick the most accurate statement referring to batch processing.

	✓
The processing of data usually takes place when the computing resources are very busy	
It is used in the control of traffic lights	
It typically involves the use of sensors and control systems	
It is typically used in the printing of electricity bills	

[1]

(b) Tick the most accurate statement referring to real time processing.

	✓
It would be suitable for controlling a car park barrier	
It does **not** require an immediate response	
It is never used 24 hours a day for the same task	
It is where a group of transactions is collected over a period of time before being processed	

[1]

Theatres often use interactive processing to allow customers to book seats.

(c) Describe what is meant by the term interactive processing.

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(d) Give reasons why interactive processing would be used instead of batch processing for booking seats in a theatre.

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2 For each of the **two** tables below, tick the most accurate statement referring to the use of simulation in pilot training.

(a)

	✓
The running and maintenance costs of a flight simulator are far greater than those of an actual aircraft	
Dangerous situations can be simulated without putting the trainee pilot's life at risk	
It is more difficult to simulate adverse weather conditions in a flight simulator	
The level of noise in a flight simulator is far louder than in an aircraft	

[1]

(b)

	✓
The pilot has to be able to use spreadsheet software efficiently	
There is no limit to how many ‘g’ a simulator can subject a pilot to, and for how long	
The simulator response will always be exactly the same as an actual aeroplane, as there are so few variables	
Sometimes pilots become overconfident after training on the simulator	

[1]

3 Pardeep has a lot of confidential data on her desktop computer which she wishes to transfer to a backing storage device. She is not sure whether to buy an extra hard disk drive or a magnetic tape drive.

Evaluate, by weighing up the advantages and disadvantages, the use of a hard disk drive compared with a magnetic tape drive.

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4 By describing RAM and ROM, explain why a computer needs both types of memory.

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5 Many businesses code data when storing information. A shop which sells shirts may wish to code the size and colour of each shirt, e.g. a yellow shirt with chest size 100 cm would be coded as 100Y.

Discuss the advantages and disadvantages of coding data.

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..... [6]

6 Many people have computers and devices in the home and in their cars that can be controlled using a gesture based interface or a dialogue interface.

Evaluate, by weighing up the advantages and disadvantages, these two user interfaces when compared to each other.

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7 Here is part of a spreadsheet showing the earnings of a selection of workers at a company.

E10 | =D10*VLOOKUP(B10,jobrate,3,0)

	A	B	C	D	E	F
1					Standard rate per hour	
2			As	Assistant	$12	
3			Se	Secretary	$20	
4			Cl	Clerk	$15	
5						
6	Worker ID	Job code	Job type	Hours worked this week	Wage paid this week	
7	42158P	As	Assistant	39	$468	
8	63119M	Se	Secretary	40	$800	
9	63214P	Se	Secretary	35	$700	
10	89261N	Cl	Clerk	36	$540	
11	97643M	As	Assistant	40	$480	

(a) The formula in cell E10 contains a named range, shaded in the spreadsheet.

Describe, step by step, how the named range was created.

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(b) Explain, step by step, what the formula in cell E10 does.

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(c) Give **two** reasons why a named range is used.

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[2]

(d) Explain, step by step, how to perform a filter on secretaries earning more than $750.

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Here is the same spreadsheet showing the formula in cell C7, **VLOOKUP(B7,C$2:D$4,2,0)**.

C7 ⋮ ✗ ✔ *fx* =VLOOKUP(B7,C$2:D$4,2,0)

	A	B	C	D	E	F
1				Standard rate per hour		
2			As	Assistant	$12	
3			Se	Secretary	$20	
4			Cl	Clerk	$15	
5						
6	Worker ID	Job code	Job type	Hours worked this week	Wage paid this week	
7	42158P	As	Assistant	39	$468	
8	63119M	Se	Secretary	40	$800	
9	63214P	Se	Secretary	35	$700	
10	89261N	Cl	Clerk	36	$540	
11	97643M	As	Assistant	40	$480	

(e) By explaining what each part of the formula does, describe how it displays 'Assistant' in C7.

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8 Describe the features of the encryption protocols TLS and SSL.

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9 Data on a CD ROM is regarded as static data.

Using another appropriate example, describe the characteristics of static data.

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10 In a recent BBC news bulletin it was said:

'*It takes 82 seconds for cyber-thieves to ensnare the first victim of a phishing campaign, a report suggests*.'

'*Hackers exploit router flaws in unusual pharming attack*.'

Describe what is meant by:

(a) phishing

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(b) pharming.

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11 Explain why generic file formats are needed when using word processing software and spreadsheet software.

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12 It is possible to use sound editing software to vary the sampling rate in an audio file.

Describe what is meant by sampling, sampling rate and the factors that should be considered when varying the sampling rate.

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13 Gurjeet told his classmate Sunil “I’m going to search the internet to help me with my homework”. Sunil said “Surely you mean you are going to search the world wide web!”

Explain, in detail, the differences between the internet and the world wide web.

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INFORMATION TECHNOLOGY **9626/12**

Paper 1 Theory **February/March 2021**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
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INFORMATION

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This document has **16** pages.

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[Turn over

1 There are many internal hardware devices in a typical personal computer.

(a) Tick the most accurate statement referring to hardware.

	✓
There are two types of hardware: application hardware and system hardware.	
Hardware consists of the programs used to direct the operation of a computer.	
Hardware is the collection of physical components that constitute a computer system.	
A device driver is an example of an item of hardware.	

[1]

(b) RAM is an example of an item of internal hardware.

Tick the most accurate statement referring to RAM.

	✓
A hard disk drive allows quicker access to data than RAM.	
RAM is portable and very easy to move from device to device.	
RAM stores active program data.	
Per unit of memory, RAM is far cheaper to buy than a hard disk drive.	

[1]

(c) **(i)** Describe the components of the central processing unit and describe their function.

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(ii) A computer contains a number of internal hardware components.

Describe the following:

Motherboard .....

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Sound card .....

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[4]

2 **(a)** Tick the most accurate statement referring to a gesture-based interface.

	✓
It is easier for users with physical disabilities to use a keyboard or mouse than make gestures.	
It is easier to enter text with a gesture-based interface than with a command line interface.	
Gestures are more accurately interpreted than mouse clicks.	
Some gestures may be unintentional but still initiate a reaction from the device.	

[1]

(b) Tick the most accurate statement referring to a dialogue interface.

	✓
Dialogue interfaces are more difficult to use than command line interfaces.	
A dialogue interface uses less memory than a command line interface.	
A dialogue interface may not operate reliably if there is background noise.	
Dialogue interfaces have unlimited vocabularies.	

[1]

(c) Many software developers use a command line interface (CLI).

Evaluate, by weighing up the advantages and disadvantages, a CLI compared to a graphical user interface (GUI).

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3 A printed newspaper is regarded as a source of static data.

(a) Give **two** other examples of sources of static data.

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(b) Explain why a printed newspaper is regarded as a source of static data.

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4 A sports club secretary has entered data from a source document into a database on his own computer. The original document contains correct data but he made some errors when copying the data into the database. Membership number is text data type and Membership fee paid is currency.

Membership number	Date of birth	Membership fee paid ($)
45237	13/04/1996	35
34859	09/12/1998	46
28695	12/11/1992	43
61892	01/02/2001	12
56123	27/06/1999	15

Source document

Membership number	Date of birth	Membership fee paid ($)
45237	13/04/1996	35
3485	09/12/1998	46
28695	12/12/1992	43
61892	01/02/2001	12
56123	27/06/1999	5

Database

The **three** errors were:

15 has been copied as 5
34859 has been copied as 3485
12/11/1992 has been copied as 12/12/1992

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(a) (i) Explain, in detail, why one of the errors could **only** have been detected using verification.

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(ii) Describe the most appropriate method of verification that could be used to trap this error.

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(b) For each of the other **two** errors, describe the most appropriate validation check which would have trapped the error.

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[2]

5 Joan has written some gaming software in a high-level programming language. She intends to publish and sell copies of the game. She wants to test it on her computer. The software will be converted into a lower-level language.

Evaluate, by weighing up the advantages and disadvantages, the use of a compiler for this purpose.

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6 Some people use modern technology in their everyday lives. Explain how aspects of such modern technology influence the digital divide.

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7 Explain the relationship between the World Wide Web and the internet.

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8 Explain how the reporting feature of a management information system (MIS) helps an organisation in the running of its business.

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9 Describe the format of a mobile phone (cell) network and how it enables the transmission of voice and data.

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10 Analyse the use of real-time processing when used in the computer control of car park barriers.

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11 When creating queries in a database, a function IIf() is often used. IIf is equivalent to the IF function found in spreadsheet software.

Examine the following database query.

IIf([Examdate]>Date(),"not yet",IIf([Examdate]=Date(),"today's the day","you've missed it"))

Date() returns the current date. [Examdate] is written as "Examdate" in some software.

Describe, in detail, what the database query does.

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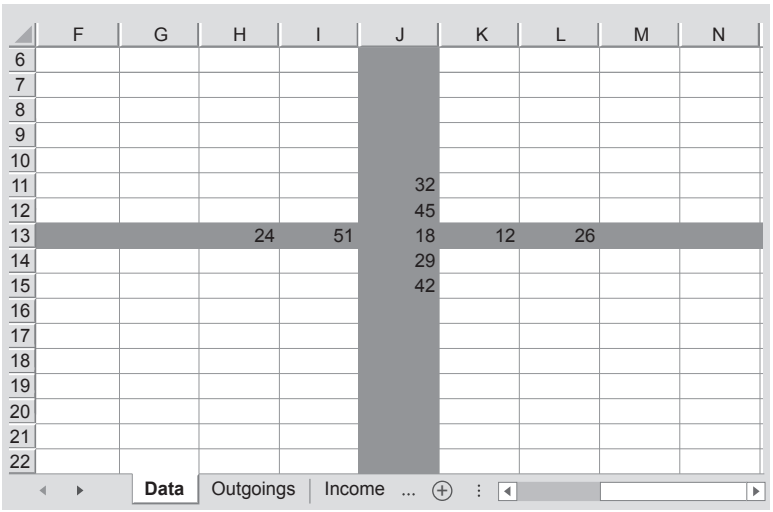
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12



(a) Explain, using examples from the spreadsheet above and its contents, what is meant by:

(i) cells,

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(ii) rows,

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(iii) columns,

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(iv) worksheets.

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(b) Apart from their structure, explain why spreadsheets are often used to model household finances.

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Please turn over for Question 13.

13 A company stores records of its employees in alphabetical order of name.

Explain what is meant by the indexed sequential access method, describing how it could be used to retrieve the records of employees named Patel.

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CANDIDATE NAME

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INFORMATION TECHNOLOGY **9626/12**

Paper 1 Theory **February/March 2022**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
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INFORMATION

- The total mark for this paper is 70.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **12** pages. Any blank pages are indicated.

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[Turn over

1 A collection of data could be the following:

1989, Paolo, √, ₹, ≠, OR

(a) Explain, by defining the term 'data', why this is just a collection of data and is not information.

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(b) Select **three** of these items of data. Explain what information each item might represent and how it could be used.

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[3]

2 An interpreter is a type of system software.

Describe how an interpreter operates.

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..... [4]

3 Mainframe computers are particularly suited to carrying out the transaction processing of online banks.

(a) Describe the features of a transaction processing system.

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(b) Discuss the advantages and disadvantages of the use of mainframe computers.

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4 Computerised weather stations are used to monitor a number of analogue atmospheric variables.

Describe how **three** of these variables are collected and what needs to happen to the data before it can be used by a computer.

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5 Some expert systems use a mixture of forward chaining and backward chaining. Some use forward chaining only.

Describe the process of forward chaining.

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6 A school secretary is updating a database by copying data from some hard copy records.
Here are three of the records:

First_name	Family_name	Student_number	Previous_school	Date_of_birth
Jonathon	Odoki	12036	St Mary's	07/12/2004
Sven	Johansson	09132	The Grange	13/10/2007
Manjit	Sangherra	10102	Bluestars	11/03/2007

The Student_number will be stored as text, with the first two characters representing the year group the student belongs to.

The Student_number and Date_of_birth are the only data that need to be validated as they are entered. A range check, limit check or presence check would not be appropriate for validating these data.

(a) Describe, in precise detail, **three** appropriate validation checks that could be used to validate the data. Each validation check must be different.

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[6]

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(b) As well as validation, verification will need to be carried out.

Explain why verification is needed in addition to validation. Use examples from your answers to part **(a)**.

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7 The average (mean) of a set of numbers can be found by adding the numbers together and dividing the resulting total by how many numbers there are in the set.

Write an algorithm in pseudocode to enter and find the average of a set of numbers using a REPEAT…UNTIL loop. Your algorithm must work for different sets of numbers.

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8 Phishing is an attempt by fraudsters to gather personal information using email. Receivers of emails should use several different methods to prevent phishing attacks.

Evaluate the different methods which can be used to prevent phishing.

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9 Hackers sometimes commit fraud after causing spyware to be installed on other users' computers.

(a) Describe what is meant by the term 'spyware'.

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(b) Explain how a hacker uses the information gained from spyware to commit fraud.

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10 **(a)** Define the following database terms.

(i) One-to-one relationship

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..... [1]

(ii) One-to-many relationship

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..... [1]

(iii) Compound key

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..... [1]

(b) Explain how a many-to-many relationship can be created in a database.

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INFORMATION TECHNOLOGY **9626/12**

Paper 1 Theory **February/March 2023**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
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INFORMATION

- The total mark for this paper is 70.
- The number of marks for each question or part question is shown in brackets [ ].

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[Turn over

1 Data is collected and processed to become information. Many factors affect the quality of the resulting information.

(a) Describe what is meant by accurate information.

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(b) Describe how the use of interviews and questionnaires for collecting data may affect the accuracy of the resulting information.

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Online processing can be used when goods are purchased.

(c) Define what is meant by online processing.

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..... [1]

(d) Johann has previously registered with an online store.

Describe the steps he takes when purchasing goods from the online store.

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2 A linker is a type of system software used during the process of translating high level language programs to machine code.

Discuss the benefits and drawbacks of using a linker.

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3 Most computer systems have a variety of utility software including anti-virus software.

Discuss the benefits and drawbacks of anti-virus software.

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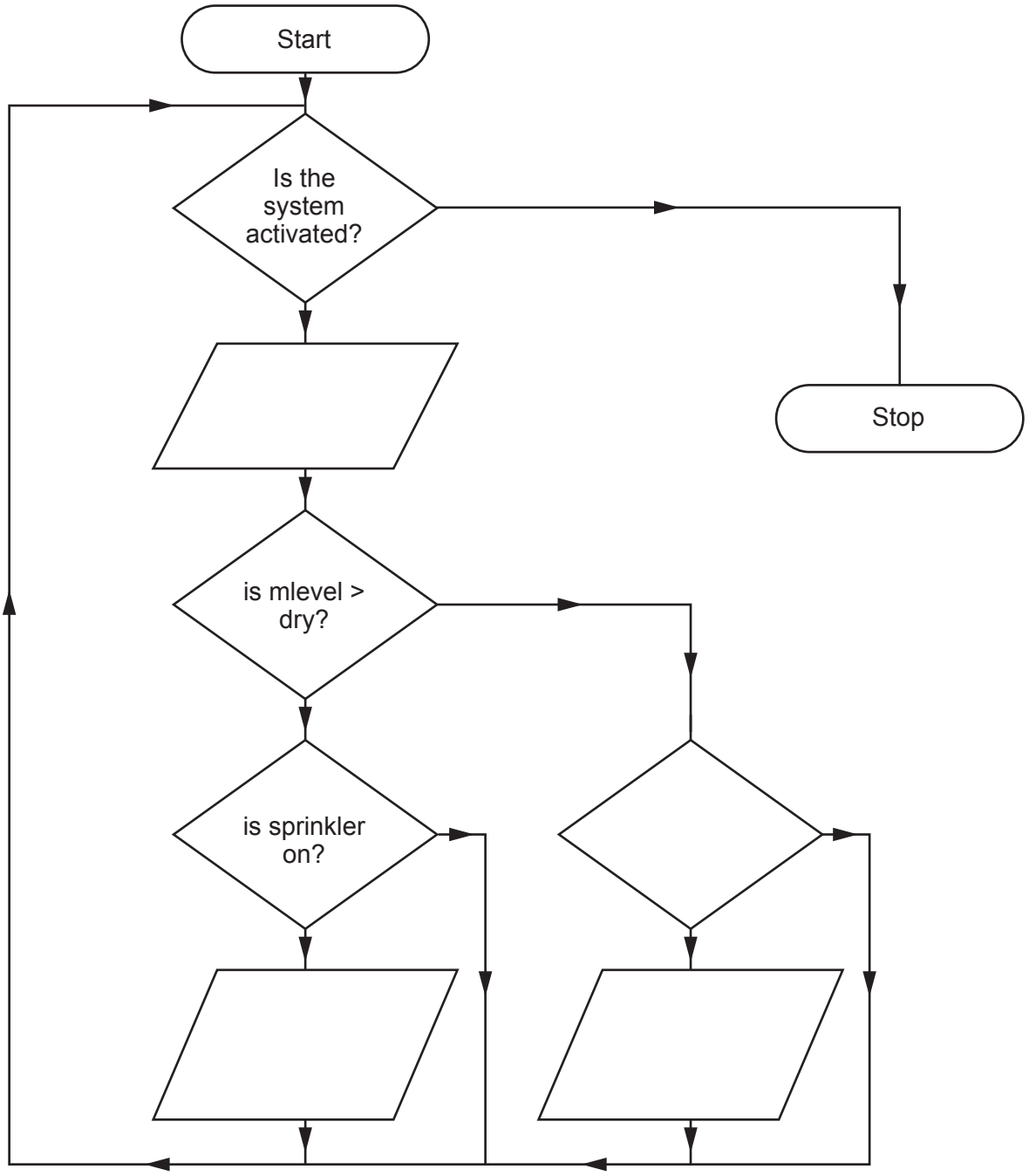
..... [6]

4 Mark has a small glasshouse. He lives in a warm country where plants can die if they are left without water for too long. The glasshouse has a microprocessor controlled sprinkler system and a moisture sensor in the soil.

Complete this flowchart to show the processing required to control the level of moisture in the soil of the glasshouse. You may assume that a pre-set value, **dry**, has already been input so that the system can test if the amount of moisture in the soil has reached or fallen below a certain level.

The user activates the microprocessor system by pressing a switch.

(a) Complete the flowchart which inputs the moisture reading in the glasshouse and switches the sprinkler on and off according to this value. It also checks if the sprinkler is already on or off before attempting to switch it on or off.



[6]

(b) Flowcharts often contain subroutines.

(i) Draw the symbol used to represent a subroutine in a flowchart.

[1]

(ii) State what you would expect to see in the first box of a subroutine.

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..... [1]

5 Explain why encryption is needed.

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..... [4]

6 Sensors are used to measure physical variables.

(a) Describe these sensors and give an appropriate use in a monitoring system for each.

(i) Light .....

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..... [2]

(ii) Temperature .....

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..... [2]

(b) Sensors need to be calibrated before use.

Evaluate the different forms of calibration.

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7 Phishing is a method used by hackers to obtain banking information from computer users.

Explain the process of phishing.

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8 Describe how people in developing countries can be disadvantaged when trying to use online shopping.

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..... [5]

9 Describe what is meant by the malware terms.

(a) worm

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(b) Trojan

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10 Describe what is meant by the term open source file format. Include in your answer an explanation of why open source file formats are needed.

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CANDIDATE NAME

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INFORMATION TECHNOLOGY **9626/11**

Paper 1 Theory **May/June 2019**

1 hour 45 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

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Answer **all** questions.

At the end of the examination, fasten all your work securely together.
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1 The coding of data refers to using M for male and F for female, for example.

Tick the **four** most accurate statements regarding the coding of data.

	✓
Data becomes easier to interpret	
Coding of value judgements is always very accurate	
The coding of data saves storage space	
Codes cannot be used in a relational database	
It speeds up data entry	
All data can be coded	
There are fewer mistakes when entering data	
Codes are not too complicated so there are never any errors on data entry	
Validation is easier to perform on data	
Coded data makes the computer slower when searching for data	

[4]

2 The management of a hotel is introducing a computer system for producing bills for their customers. They are considering which type of printer to use.

Tick the **four** most accurate statements regarding different types of printer, if used in the hotel reception area.

	✓
Laser printers produce very high quality output	
A dot matrix printer would produce higher quality output than an inkjet printer	
An inkjet printer would be slower at producing output than a dot matrix printer	
Dot matrix printers use continuous stationery	
Dot matrix printers need the ink cartridge changing more frequently than inkjet printers	
Dot matrix printers are very quiet which is important in a hotel reception area	
A laser printer can operate in harsh conditions, unlike a dot matrix printer	
A laser printer toner cartridge is more expensive to buy than an inkjet cartridge	
Multi-part stationery is easier to use with laser printers than with dot matrix printers	
Inkjet printers are cheaper to buy than dot matrix printers	

[4]

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3 Items of data could be the following:

45612, fred, π, £, ≥ or 

Explain, by describing what is meant by information, why these are just items of data.

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.....[6]

4 Most schools have introduced e-safety lessons because teachers are becoming concerned about the way their students use the internet when not in school. One of their concerns is that students may be revealing personal information about themselves.

Describe the problems associated with this and other problems that arise from students using the internet.

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.....[6]

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5 Many companies with different branches have Local Area Networks (LANs) which connect to form a Wide Area Network (WAN).

Compare and contrast the characteristics of a LAN and a WAN owned by one company.

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.....[6]

6 Louis has partly typed up the notes from his IT lesson. He wants to complete these at home. Sophia, a fellow student, tells him that it may be difficult as he has been using proprietary software.

Explain what is meant by proprietary and open-source file formats, giving reasons why open-source file formats are needed.

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7 An operating system is system software that manages computer hardware and software resources.

Describe the tasks that an operating system carries out.

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.....[4]

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8 There are a number of ways that fraudsters can obtain money from customers' bank accounts.

(a) Describe how the fraudster would use vishing to obtain a customer's personal data.

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(b) Describe how the fraudster would use smishing to obtain a customer's personal data.

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.....[3]

9 Paolo is the owner of a small company with ten workers. He is developing a spreadsheet to calculate the payroll. He has started by importing a text file of the workers' names. The list consisted of each worker's family name followed by a comma then a space then their first name.

He wants to split the names into separate family names and first names by finding the position of the comma in each cell in column A.

He has discovered a new function, LOCATE, which gives the position of the first occurrence of a single character in a word.
For example, =LOCATE("e",I4) gives the result 2.

His first attempt at the spreadsheet is shown below.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Name		Position of comma		Family name	First name			Job	Weekly pay		Rate per hour	Hours worked
2													
3	Smith, John								Polishing	450		15	30
4	McDonnell, Louise								Secretary	500		10	50
5	Hernandes, Juan								Welding	450		10	45
6	Kennedy, Sean								Orders	500		16	31.25
7	Gustaffsson, Johan								Fitter	500		20	25
8	Davies, Iueuan								Electrician	550		11	50
9	Lopez, Ramon								Manager	500		25	20
10	Garcia, Isabel								Painting	550		20	27.5
11	Costa, Albertina								Invoices	450		15	30
12	Fernandez, Mariela								Sales	550		11	50
13													

(a) Using this LOCATE function, write down the formula he would use in cell C3. The formula must be easy to replicate.

= .....[2]

You can use the space below for any working you need.

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(b) He will use a formula in cell E3 to produce Smith. Making use of cell C3, write down the formula, **without** using the MID function, which must be easy to replicate.

There must be no spaces, before, or after the name.

= .....[3]

You can use the space below for any working you need.

(c) He will use a formula in cell F3 to produce John. Making use of cell C3, write down the formula, **without** using the MID function, which must be easy to replicate.

There must be no spaces before, or after, the name.

.....[3]

You can use the space below for any working you need.

(d) Describe the steps he would use to format the Weekly pay column so that the only difference that would be made would be to place a $ sign in front of the value.

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(e) Using **only** the sort feature on columns I and J, explain how you would be able to get the following values in the given cells. You need to ensure data integrity is maintained.

(i) 500 in cell J6 with Secretary in cell I6 and Davies, Iueuan in A12.

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.....[3]

(ii) 500 in J7 with Orders in I8 and Smith, John in A11.

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10 A car park has a barrier which operates automatically. It opens when a car approaches it and closes when the car is clear of the barrier.

Describe, including the use of sensors, how a microprocessor controls this operation.

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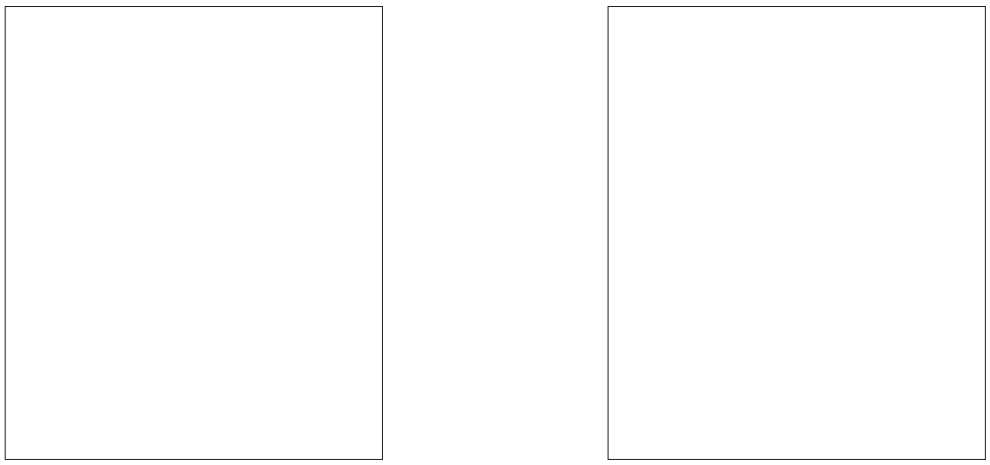
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.....[4]

11 Maria, the A level students' secretary, has been given the task of creating a database. It will store the records of each subject taught and the records of the teachers involved. She has decided to produce a relational database.
The **Teacher** records will have the fields **Teacher_ID**, **Family_name**, **First_name**, **Subject_ID** and **Number_of_classes**. The **Subject** records will have the fields **Subject_code**, **Subject_name**, **Subject_level** and **Number_of_lessons**.

(a) Complete the diagram below to show how the tables will be combined.



[4]

(b) Using the design of your database from part **(a)**, describe how you would create it.

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.....[4]

12 A government database stores the licence plates of cars. In this particular country they all follow the same pattern, e.g.

BG136KG
CD125FH
FT342LM
NQ439SR

Apart from a presence check, describe **two** appropriate validation checks which could be applied to licence plates. For each check give an appropriate example of invalid data that would be rejected. Each example must be different.

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13 IT professionals often use command line interfaces (CLIs) to communicate with computers. Office workers, however, prefer graphical user interfaces (GUIs). In the home, gesture based interfaces (GBIs) are increasingly being used.

Evaluate these three types of interface.

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14 Many students use the internet in school to search for information to help with their classwork.

Apart from e-safety, discuss the benefits and drawbacks of students using the internet for this purpose.

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0771104865

INFORMATION TECHNOLOGY **9626/12**

Paper 1 Theory **May/June 2019**

1 hour 45 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

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Answer **all** questions.

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1 Tick the **four** statements referring to asymmetric encryption which are true.

	✓
It is often referred to as public key encryption	
It uses a pair of keys, a public key and a private key	
The public key and the private key are published to everyone who wants to send a message	
Anyone with a copy of the public key can read encrypted data	
It is possible to deduce the private key from the public key	
SSL is a protocol that uses asymmetric encryption	
Keys used in symmetric encryption are longer, compared to asymmetric keys	
Asymmetric encryption is slower to convert than symmetric encryption and requires far more processing power	
Digital certificates are not used with asymmetric encryption	
The use of an asymmetric key algorithm always ensures security of a message	

[4]

2 Tick the **four** statements referring to malware which are true.

	✓
A virus corrupts files but does not delete them	
Ransomware continually deletes files until a ransom is paid	
One purpose of releasing a worm is to occupy as much bandwidth in a network as possible	
Ransomware is often initiated by means of a trojan horse	
A trojan horse is a type of malware within an email attachment that is downloaded without the user understanding the consequences	
Spyware is a type of software that removes key loggers from a hard disk	
Adware is additional software attached to a virus when it is downloaded	
Malicious bots are never used to instigate a SPAM attack	
A rootkit enables a hacker to gain administrator rights in a victim's computer	
A trojan horse just enables a virus to be transmitted. It never deletes data itself	

[4]

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3 People in a village are aware that the local river is being polluted. They are concerned that this is being caused by a nearby factory.

Describe how computers and sensors could be used to monitor the level of pollution in the river in order to determine whether the factory is responsible for the pollution.

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.....[5]

4 Some people get confused when trying to establish what is data, information or knowledge. The sequence of numbers 192.168.1.254 could be an example of data.

Using this example, or another of your choice, explain how data, information and knowledge are linked.

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.....[4]

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5 A large company with branches in many countries has decided to install a Virtual Private Network (VPN).

(a) Describe the purpose of such a network.

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.....[2]

(b) Explain how it achieves this purpose.

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.....[4]

6 Video-conferencing is a commonly used method of communication. It has affected different sections of society.

Describe **two** effects of video-conferencing for each of the following groups of people:

Legislators (government members who make legislation or laws)

1 .....

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2 .....

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Business people

1 .....

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2 .....

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Television news presenters

1 .....

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[6]

7 Relational databases can have many components and properties.

(a) Describe what is meant by the following terms:

(i) Primary key .....

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(ii) Foreign key .....

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(iii) Compound key .....

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.....[3]

(b) Explain why referential integrity is important in a relational database.

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.....[3]

8 The owners of a bank are considering changing its system of reading cheques from Magnetic Ink Character Recognition (MICR) to Optical Character Recognition (OCR).

Give **two** advantages and **two** disadvantages of using OCR instead of MICR.

Advantage 1 .....

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Advantage 2 .....

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Disadvantage 1 .....

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Disadvantage 2 .....

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.....[4]

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9 An expert system consists of many components.

Describe how the knowledge base is used to interact with the other components when an expert system is used to solve a problem.

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10 An examinations officer has created a spreadsheet containing a list of students and their dates of birth. It will be used to calculate each student's age in years. This is the age that they will be on the date of a science exam which is on 21/05/2019.

	A	B	C	D	E	F	G	H
1								
2	Family name	First name	Date of birth (dd/mm/yyyy)	Age	Number of exams entered for		21/05/2019	
3								
4	Purewal	Sandeep	17/05/2002	17	4		6	number of students aged 17 taking 3 exams
5	Jones	Ieuan	15/10/2001	17	3		4	number of students aged 16 taking 4 exams
6	Astle	Roberta	21/07/2002	16	4			
7	Herreira	David	10/08/2002	16	4			
8	Maninga	Jeff	17/09/2001	17	3			
9	Charlton	Leona	08/03/2002	17	4			
10	van Gaal	Ruud	09/02/2002	17	4			
11	Alonso	Maria	17/11/2001	17	3			
12	Lagat	Mary	10/11/2001	17	3			
13	Kibaki	Robert	13/04/2002	17	4			
14	Mbabasi	Apolo	30/06/2002	16	5			
15	Dansua	Albert	24/06/2002	16	4			
16	Kirshwan	Kiran	30/08/2002	16	3			
17	Patel	Alpa	15/03/2002	17	4			
18	Bhutto	Waseem	14/09/2001	17	3			
19	Abbasi	Akhtar	18/12/2001	17	4			
20	Kunwar	Ishwar	16/01/2002	17	3			
21	al-Hafi	Saad	14/03/2001	18	4			
22	Hala	Bassem	25/05/2002	16	5			
23	Namet	Fawzi	31/05/2002	16	4			

(a) She entered a formula in D4 to calculate the student's age.

Write down the formula she used. This formula should be easy to replicate.

You can assume for the purpose of this exercise that the number of days in a year averages out to 365.25.

= .....[4]

You can use the space below for any working you need.

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(b) (i) The spreadsheet is to be sorted in ascending order of *Age* and then descending order of *Number of exams entered for*.

Write down the values which would be displayed in D9 and E9.

.....[2]

You can use the space below for any working you need.

(ii) The **original** spreadsheet is to be sorted in descending order of *Age* and then ascending order of *Number of exams entered for*.

Write down the values which would be displayed in D17 and E17.

.....[2]

You can use the space below for any working you need.

(c) A teacher wanted to know the number of students aged 17 taking three exams.

Write down the formula containing a counting function she used in cell G4 to calculate this number. This formula should work even if the data changes.

= .....[5]

You can use the space below for any working you need.

(d) The teacher now wants to know the number of students aged 16 taking four exams.

Write down the formula containing a counting function she used in cell G5 to calculate this number. This formula should work even if the data changes.

.....[3]

You can use the space below for any working you need.

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11 A library database consists of one table containing details of all the books in a library and another table containing details of all the borrowers. Parts of the tables are shown below.

Books table

ISBN	Title	Author	Publisher	ReplacementCost
0099445530	The Flamboya Tree	Clara Olink Kelly	Bow Books	€7.99
0099460874	I Capture the Castle	Dodie Smith	Archaic	€5.99
0439977797	The Tin Princess	Philip Pullman	Panda Books	€5.99
0553812645	Ten Thousand Sorrows	Elizabeth Kim	Welter Books	€6.99
0684817306	Kitchen Privileges	Mary Higgins Clark	Grimon & Custer	€12.99
1857028899	The Code Book	Simon Singh	Quatre Domaine	€8.99
9780593057063	61 Hours	Lee Child	Crossearth Publishers	€18.99
9780593065709	The Affair	Lee Child	Crossearth Publishers	€18.99
9780593072493	Inferno	Dan Brown	Crossearth Publishers	€20.00
9780752860558	Sepulchre	Kate Mosse	Nebula Books	€18.99
9781780892665	Cross Justice	James Patterson	Panda Books	€20.00
9781846558597	Midnight Sun	Jo Nesbo	Panda Books	€9.99

Borrowers table

ID	FamilyName	FirstName	Mobilephone	Residence
1	Gale	Henry	07700 900102	Midtown
2	Brown	Kurtis	01134 960675	Cheswick
3	Leadbetter	Clementine	01144 960975	Chorlton
4	Hull	Christine	07700 900643	Denby
5	Greenhalgh	Karla	08081 570372	Portlarne
6	Pooley	Fred	09098 790876	Midtown
7	Chalmers	Ben	01154 960987	Denby
8	Lewis	Christopher	07700 900152	Denby
9	Patel	Rio	08081 570976	Chorlton
10	Gopaul	Ruksana	03069 990967	Cheswick

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(a) You have been asked to create a query which will output only the names and phone numbers of all the borrowers who have a family name beginning with G and also live in Midtown.

Describe the steps you would have to follow in order to create this query, without using a query wizard or a parameter query.

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(b) You have been asked to create a query which will output only the titles and authors of all the books published by Panda Books which have a replacement cost of less than €20.

(i) Describe the steps you would have to follow in order to create this query, without using a query wizard or a parameter query.

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(ii) Describe how you would set up the query in **(b)(i)** as a parameter query.

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(c) Explain why the above tables (on page 14) do not constitute a relational database and what would need to be added to make them so.

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.....[3]

12 Validation is often used to check that data entered into a computer is sensible. A customer orders database contains details of the Customer ID number, the value of the order placed and the date the order was placed. Any new order, when it is entered, needs to be validated.

Using these examples of database fields, analyse the use of the following different validation rules – range check, type check, length check and format check.

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13 Many driving schools are asking IT companies to develop simulators for teaching customers to drive.

Evaluate the development of simulators for this purpose.

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CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

7811632120

INFORMATION TECHNOLOGY **9626/13**

Paper 1 Theory **May/June 2019**

1 hour 45 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Calculators must not be used on this paper.

Any businesses described in this paper are entirely fictitious.

DC (SC/TP) 163073/3
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1 Tick the **four** most accurate statements regarding solid state drives (SSDs), hard disk drives (HDDs), magnetic tape drives and their associated media.

	✓
Hard disks are metal platters with a magnetic coating that stores data	
Hard disk drives have no moving parts and so are not at risk of being damaged	
Magnetic tape drives can have almost instantaneous data access	
SSDs have a faster data transfer rate than magnetic tape drives	
Magnetic tape uses direct access to search for data	
Magnetic tapes are used in laptop computers more than hard disks	
SSDs store considerably more data than magnetic tapes	
SSDs still cost more per gigabyte than hard disk drives	
SSDs make more noise than the sound of hard disk drives when in use	
Magnetic tapes are used to store backups of data on file servers	

[4]

2 Tick the **four** most accurate statements regarding control and monitoring systems and the use of sensors.

	✓
A temperature sensor is able to directly control the temperature of water in a washing machine	
A pressure sensor is often used to monitor the amount of pollution in a river	
A moisture sensor is used to monitor the pollution in a river	
A humidity sensor is often used in a computerised weather station	
Microprocessors are unable to directly read the analogue data produced by a sensor	
In order to control physical variables, microprocessors send signals to actuators to take action	
Passive sensors produce both input and output	
A weather station is an example of a control system	
An air conditioning system is an example of a control system where the output affects the input	
Monitoring systems never need sensors to input data	

[4]

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3 Using a printed newspaper as an example, describe what is meant by static data.

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4 The quality of a sound recording when processed by audio editing software depends on the sampling resolution of the recording.

Describe, using examples, what is meant by sampling resolution and explain how it affects audio file sizes.

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5 A headteacher is updating the school's record-keeping system. For the new system he is considering using off-the-shelf software. When the new system is running the secretary will need to enter all the students' data. This will require accurate entry of the data.

(a) Define what is meant by the term 'software' and describe, giving examples of each, application and system software.

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(b) Describe the benefits of using off-the-shelf software.

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(c) Describe the drawbacks of using off-the-shelf software.

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(d) Describe the different methods of verification the secretary could use.

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6 Spreadsheets are commonly used to create models of real situations.

Using examples, explain why the characteristics of spreadsheets are suitable for producing financial models.

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7 The encoding of data can often require the use of codecs.

Describe the features of this type of encoding.

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8 A car hire company keeps its records in a database consisting of two tables. Parts of these tables are shown below.

Customer Table

CustomerID	FamilyName	FirstName	Phone	Residence
11106	Akula	Angela	01164 960567	Midtown
12481	Onyancha	Udoka	08081 960965	Cheswick
12675	Smith	Wesley	07700 900863	Portlarne
12934	Hall	Lauren	01134 960675	Chorlton
12943	Green	Billy	01164 960001	Midtown
13296	Jones	Trevor	01134 990694	Chorlton
13478	Weston	Wally	07700 900900	Portlarne
13542	Gunn	Jasmine	08081 570945	Cheswick
14632	Wells	Brian	09098 790142	Larrowby
15692	Williams	Ieuan	01164 960836	Midtown
15732	Vercoe	John	09098 980487	Larrowby

Car Table

ID	Make	Model	Engine	Licence	Price	Rental	CustId
1	Olep	Victor	1.8	VSE 648	€19,000	€54	12481
2	Olep	Victor	1.8	VTF 384	€19,000	€54	13478
3	Olep	Fariza	2	BFK 297	€29,000	€63	15692
4	Folkwagon	Cricket	1.4	SB A5526	€15,000	€43	14632
5	Folkwagon	Cricket	1.4	SA V4428	€15,000	€43	13296
6	Folkwagon	Swim	1.2	DD B4978	€11,000	€37	15732
7	Folkwagon	Swim	1.2	DD C8634	€11,000	€37	12934
8	Folkwagon	Piano	1.6	B G8347	€19,000	€49	11106
9	Ranolt	Melanie	1.4	12333 CD 33	€17,000	€41	12943
10	Ranolt	Laine	1.2	6289 XF 54	€11,000	€33	13542
11	Ranolt	Laine	1.2	6301 YU 38	€11,000	€33	12675

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(a) Describe how you would use database features to create a relational database from these two tables.

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(b) The manager is carrying out a survey of cars hired in certain areas. She wishes to know which cars with an engine size under 1.8 litres have been rented by people living in the Chorlton area. She only wishes to know the full names of the customers, their phone number and the make of the car.

Complete the grid below to show how the conditions for such a query would be created.

Field:						
Table:						
Sort:						
Show:	☐	☐	☐	☐	☐	☐
Criteria:						
or:						

[6]

(c) Explain why it would be extremely difficult to validate the Licence field. Include in your explanation, a detailed description of **three** validation checks, other than a presence check, which could normally be used on any alphanumeric field from the database shown on page 8.

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9 Below is a spreadsheet of a company's balance sheet. It shows its expenditure, income and profit for each year between 2009 and 2018.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1													
2			Expenditure										
3			Wages	Utilities	Stock		Total expenditure		Income		Profit		
4	2009		350250	54565	475623		€880,438		€925,365		€44,927		very bad year
5	2010		467032	61022	519429		€1,047,483		€1,097,652		€50,169		bad year
6	2011		476414	68124	567178		€1,111,716		€1,185,858		€74,142		bad year
7	2012		512209	75936	619224		€1,207,369		€1,342,020		€134,651		good year
8	2013		553156	84530	675954		€1,313,640		€1,518,483		€204,843		good year
9	2014		648589	93983	737790		€1,480,362		€1,717,886		€237,524		good year
10	2015		732877	104381	805191		€1,642,449		€1,943,211		€300,762		good year
11	2016		774954	115819	878658		€1,769,431		€2,197,828		€428,397		good year
12	2017		838925	128401	958737		€1,926,063		€2,485,546		€559,483		good year
13	2018		868577	142241	1046023		€2,056,841		€2,810,667		€753,826		good year
14													
15			>2011	<2017									
16													
17		Variable average	644357										
18													

(a) Write down the formula which uses a function, in cell G4, to calculate the *Total expenditure* for 2009.

= .....[1]

(b) The percentage profit is calculated by dividing the *Profit* by the *Income*. The company secretary has entered a formula in M4 which displays a message:

- very bad year, if *Income* was less than €1,000,000
- bad year, if the company's percentage profit was less than 10%
- good year, if the percentage profit was not less than 10%

Write down the formula he used in cell M4 which would be replicated in cells M5 to M13.

= .....

.....[6]

You can use the space below for any working you need.

(c) The manager wants to be able to find the *Variable average* of the wages bill for any sequence of years between 2009 and 2018.
He has set up the spreadsheet to calculate the average between 2011 and 2017.
To calculate the average between any other sequence of years he will just change the conditions >2011 and <2017.

Write down the formula he will need to use in C17 to calculate the *Variable average*.

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.....[6]

You can use the space below for any working you need.

10 Discuss the advantages and disadvantages of client-server and peer-to-peer networks.

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11 A business needs to search its database to find all the customers it has in the Chinese city of Beijing.

Analyse when static or dynamic parameters should be used in queries.

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CENTRE NUMBER

CANDIDATE NUMBER

3294028742

INFORMATION TECHNOLOGY **9626/11**

Paper 1 Theory **May/June 2020**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 90.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages. Blank pages are indicated.

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[Turn over

1 For each of the **two** tables below, tick the most accurate statement referring to the use of the internet for communication.

(a)

	✓
Instant messaging is a service similar to SMS provided by the internet	
Social networking sites allow messages to be posted, but not pictures or videos	
A blog allows anybody who reads it to also be able to edit it	
Chat rooms only allow one person to communicate with one other person	

[1]

(b)

	✓
Emails allow people to see each other when communicating	
Emails can be sent with somebody copied in without the other recipients knowing	
Emails always get an immediate response	
Instant messaging never results in the IM providers sending adverts to users	

[1]

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(c) Video-conferencing is another way of using the internet for communication.

Describe how a manager in one branch of a company would set up a video-conference to communicate with managers in other branches.

You can assume that the hardware and software required has already been purchased. In your description you will need to identify what each manager would need to do.

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2 **(a)** Tick the most accurate statement referring to the use of compilers and interpreters.

	✓
An interpreter creates a single executable file that runs directly on the CPU	
It is easier to debug compiled code rather than interpreted code	
With a compiler it is harder to protect intellectual property as machine code is easy to understand	
It is more difficult for hackers to modify compiled code	

[1]

(b) Tick the most accurate statement referring to the use of disk formatting.

	✓
Disk formatting increases the number of fragments in a hard disk	
Disk formatting never removes viruses from a hard disk	
Disk formatting increases the storage capacity of a blank disk	
Disk formatting is used to fully prepare a hard disk for initial use	

[1]

3 Static information sources and dynamic information sources are often used for different purposes.

Define what is meant by static and dynamic data. Using a school or college as an example, describe situations which could use a dynamic information source and those where a static information source could be used.

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4 The digital divide often exists between people in developed countries and those in developing countries.

Describe what is meant by the digital divide and how it affects these countries.

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..... [6]

5 A sports club keeps records in a database of its members and the individual sports they play. It has the following tables of data:

Members' names
Members' contact details
Details of when they started playing the sport
All the possible sports which can be played by the members.

Names	Contact_details	Started_sport	Sports

(a) Complete the tables shown above, writing down examples of the **field names** for **three** of the fields in each table. Indicate the key fields in each table and the relationships that exist between each adjacent table. Indicate the type of each relationship. Some tables have one relationship and others have a maximum of two relationships. [8]

(b) Two of the tables could possibly be regarded as having a many-to-many relationship.
Using the example of the tables shown above, explain what is meant by a many-to-many relationship.

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6 Many companies use batch processing for producing workers' payslips.

(a) Describe, using payroll as an example, what is meant by batch processing.

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(b) Explain why batch processing is used instead of real-time processing for an application such as payroll.

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7 Designers of user interfaces tend to base their designs on what they interpret to be a user's mental model.

Evaluate, by weighing up the advantages and disadvantages, the use of a mental model to design a user interface.

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8 Explain how encryption protects data stored on a hard disk.

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9 By explaining what the purpose of each is, describe the differences between verification and validation.

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10 A researcher is interviewing some people. To begin with, she wants to record their personal details using a six-character code. For example, 47FT12 means the person is 47 years old, in full-time employment and left school after year 12.

Explain what the code 34PT10 means and why there may be problems with coding other people in this way. Include in your explanation ways of improving this method.

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11 Mary has been put in charge of the computerised school weather station. At the moment the system has a monitor to show the outputs. She wants to replace the monitor with a printer.

By weighing up the advantages and disadvantages, evaluate the system having a monitor rather than a printer.

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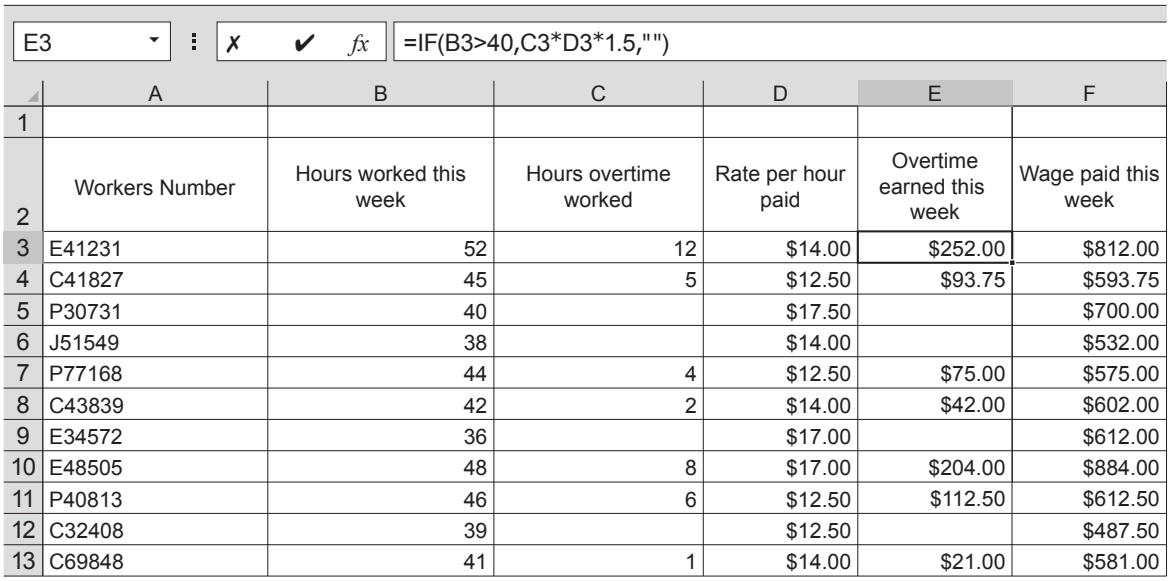
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..... [6]

12 Here is part of a spreadsheet showing the wages earned by a group of workers in a company.



E3 ⋮ ✗ ✔ *fx* =IF(B3>40,C3*D3*1.5,"")

	A	B	C	D	E	F
1						
2	Workers Number	Hours worked this week	Hours overtime worked	Rate per hour paid	Overtime earned this week	Wage paid this week
3	E41231	52	12	$14.00	$252.00	$812.00
4	C41827	45	5	$12.50	$93.75	$593.75
5	P30731	40		$17.50		$700.00
6	J51549	38		$14.00		$532.00
7	P77168	44	4	$12.50	$75.00	$575.00
8	C43839	42	2	$14.00	$42.00	$602.00
9	E34572	36		$17.00		$612.00
10	E48505	48	8	$17.00	$204.00	$884.00
11	P40813	46	6	$12.50	$112.50	$612.50
12	C32408	39		$12.50		$487.50
13	C69848	41	1	$14.00	$21.00	$581.00

(a) Without referring to other cell references, explain, in detail, what every part of the formulae in cells E3 and E5 do. You may assume that the formula has been replicated down for all workers.

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(b) This spreadsheet has not been tested.

Describe the test plan you would use to make sure there are no errors in the formulae in column E of this spreadsheet. You can assume that the formula in column C works perfectly and does not need changing.

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(c) Using **only** the sort feature explain how you would be able to make the spreadsheet used in part **(a)** produce this display.

	A	B	C	D	E	F
1						
2	Workers Number	Hours worked this week	Hours overtime worked	Rate per hour paid	Overtime earned this week	Wage paid this week
3	P40813	46	6	$12.50	$112.50	$612.50
4	C41827	45	5	$12.50	$93.75	$593.75
5	P77168	44	4	$12.50	$75.00	$575.00
6	C32408	39		$12.50		$487.50
7	E41231	52	12	$14.00	$252.00	$812.00
8	C43839	42	2	$14.00	$42.00	$602.00
9	C69848	41	1	$14.00	$21.00	$581.00
10	J51549	38		$14.00		$532.00
11	E48505	48	8	$17.00	$204.00	$884.00
12	E34572	36		$17.00		$612.00
13	P30731	40		$17.50		$700.00

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[Turn over

13 Computers are often used for environmental monitoring applications such as pollution in rivers.

Evaluate, by weighing up the advantages and disadvantages, the use of computers rather than humans in such scenarios.

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0538073010

INFORMATION TECHNOLOGY **9626/12**

Paper 1 Theory **May/June 2020**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 90.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages. Blank pages are indicated.

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1 For each of the **two** tables below, tick the most accurate statement referring to symmetric encryption.

(a)

	✓
Symmetric encryption is a newer method of encryption compared to asymmetric encryption	
With symmetric encryption the public key is used to encrypt the data	
With symmetric encryption you have to use the same key to encrypt every message	
Symmetric encryption only uses a single private key	

[1]

(b)

	✓
With symmetric encryption the private key must be kept private by both the sender and the receiver	
Symmetric encryption is often referred to as public key encryption	
It is possible to deduce the private key from the public key	
With symmetric encryption anyone with a copy of the public key can encrypt information	

[1]

2 **(a)** Tick the most accurate statement regarding the use of word processing software for producing documents.

	✓
It is easier to format paragraphs compared to database software	
It is easier to manipulate statistics compared to using spreadsheet software	
Word processing packages can be used to produce all multimedia features of a website	
Databases are more difficult to filter than word processed documents	

[1]

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(b) Tick the most accurate statement regarding the use of other software to manipulate data.

	✓
It is easier to create a query using database software rather than using spreadsheet software	
It is easier to create a report using spreadsheet software rather than using database software	
It is harder to enter data and produce charts showing sales growth with a spreadsheet than using video editing software	
The easiest way to trim a sound sample is by using photo editing software	

[1]

(c) Evaluate, by weighing up the advantages and disadvantages, the use of a web authoring package rather than a text editor to create a web page.

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3 Using numerical examples, describe what is meant by the following terms.

Data .....

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Information .....

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Knowledge .....

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[6]

4 Several radio programmes have recently highlighted e-safety as an issue that schools need to address.

Describe the advantages of a school having an e-safety policy.

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..... [6]

5 Many schools in the more remote areas of Kenya want to install video-conferencing equipment. This is so their students can be in touch with students from other Kenyan schools.

(a) Discuss the impact of video-conferencing on the more remote schools and their students, compared with having to visit the other schools to stay in touch.

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..... [6]

Several technologies can be used to deliver video-conferencing.

Explain what is meant by the following terms.

(b) ADSL

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..... [3]

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(c) IEEE 802.11

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..... [3]

6 Personal data and other sensitive types of data often need to be kept confidential or restricted to authorised staff.

Explain why personal data should be kept confidential.

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7 Here is part of a database showing the earnings of a group of workers in a company.

WorkerID	Job_type	Hours_worked	Hourly_rate	Weekly_pay
56321	Tyre changer	42	€17.00	€714.00
58191	Foreman	40	€22.00	€880.00
58694	Mechanic	38	€20.00	€760.00
59676	Painter	37	€18.00	€666.00
61230	Sander	39	€15.00	€585.00
63214	Sander	36	€15.00	€540.00
63295	Polisher	38	€14.00	€532.00
64387	Polisher	40	€14.00	€560.00
67152	Mechanic	37	€20.00	€740.00
67328	Painter	36	€18.00	€648.00
68947	Painter	38	€18.00	€684.00

(a) Describe, using an example from the database, the features of a calculated field.

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(b) A number of fields in the database are not shown. Examples of these are first name and family name. In this database the WorkerID is a primary key.

Explain why it is good practice to have a primary key in this database.

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(c) Describe what a consistency validation check is and explain how it could be used in this database as a check when inputting data.

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8 Describe how **two** different safety issues can arise from the use of IT. For each issue give **one** method of helping to prevent it. Each method must be different.

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9 Check-out terminals in supermarkets have a number of different input devices for entering data about products and for payment purposes.

Analyse the use of these input devices in the supermarket.

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10 Ayesha is searching a school database for all the students studying biology. She has used a static parameter query but has been advised to set the query up as a dynamic parameter query.

Describe what is meant by a dynamic parameter query and give the advantages of using one instead of a static parameter query.

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..... [5]

11 A construction company is preparing to build a road bridge. Before building the bridge, it will create a computer model of the bridge to test the designs that have been produced.

Explain why the company considers it necessary to produce a computer model.

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12 A headteacher wishes to buy software which will help him manage the finances of the school. When considering which software to buy he has to choose between custom written software and off-the-shelf software.

Describe the differences between these two types of software.

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13 A company has decided to develop its computer network into an extranet.

Describe the purpose of a company extranet.

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..... [5]

14 The inference engine within an expert system uses a form of reasoning.

Explain what this form of reasoning consists of.

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4445188864

INFORMATION TECHNOLOGY **9626/13**

Paper 1 Theory **May/June 2020**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 90.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages. Blank pages are indicated.

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1 **(a)** Tick the most accurate statement referring to the internet.

	✓
The internet is a collection of web pages which can be accessed using a web browser	
The internet is the global system of interconnected computer networks	
The internet is only accessed using a switch	
The internet is an internal restricted access network	

[1]

(b) Tick the most accurate statement referring to the services the internet provides.

	✓
Word processing is a service only accessed using the internet	
Spreadsheet software is a service only accessed using the internet	
The World Wide Web is a service only accessed using the internet	
Email is a service only accessed using the internet	

[1]

(c) The internet is a type of WAN. Describe what is meant by a WAN.

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..... [4]

2 **(a)** Tick the most accurate statement referring to video files.

	✓
It is only possible to export a video file using the file format it has been created in	
Video files are often stored in jpeg format	
It is possible to convert a video file into a gif format	
It is **not** possible to insert a still image into a video file	

[1]

(b) Tick the most accurate statement referring to the use of features of video editing software.

	✓
It is **not** possible to alter the speed of a video once it has been saved	
Trimming a video clip is achieved by removing unwanted footage	
It is **not** possible to add sound to a video once the video is finished	
An aspect ratio is the number of frames in a section of a video compared to the total number of frames	

[1]

3 BNMA, a large banking organisation, uses management information system (MIS) software in its day to day business concerns.

Without referring to what MIS consists of, or to the specific role of the MIS manager, explain how BNMA could use this MIS software.

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4 Chung and Fai both work at the Vacation Hotel. Chung told Fai that the hotel had just installed a car park barrier system to prevent local shoppers from using the car park. He told her that the manager had said that it was a computer-controlled system. Pressure sensors have not been used and it is impossible for the barrier to lower itself on to a car.

Describe, in detail, how such a system would operate both for a guest arriving at the hotel and for a guest leaving the hotel.

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5 Companies often carry out investigations into the public's likes and dislikes. Investigations such as this can be done using direct sources or indirect sources of information.

(a) Describe the advantages of gathering data from indirect data sources.

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(b) Describe the disadvantages of gathering data from indirect data sources.

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6 Joanne owns a store which sells films as DVD and blu-ray discs. Each disc has a six-character code number and each customer has an eight-character customer number. She has developed a database system which contains information about the films. Each record contains the following data:

Film title, director, code_number, DVD (Y/N), date_of_sale, customer_number.

Describe the features of an appropriate online data entry form for one film which would, where appropriate, allow the user to enter data by clicking the mouse rather than entering text.

Do **not** draw the screen and data entry form in your description.

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7 A program written in a simple high level programming language has a mistake in it.

Line 20 should say LET A = X + Y, but the program was typed as:

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10  INPUT X, Y
20  LE A = X + Y
30  PRINT X, " + ",Y, " = ", A
40  END
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Describe the **two** ways the program, including the error, could be translated so that it runs on a computer.

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8 The North Indonesian Motor Company makes cars in the city of Jakarta. It has a large factory producing the components, for the cars in the city of Surabaya. Dian is the director of communications in Jakarta. He has been asked by the other directors to set up a video-conference so that they can communicate with the directors based at Surabaya.

The directors in both cities currently use computers in a LAN to carry out their work.

(a) Give reasons why the current arrangement of computers would not enable Dian to set up a video-conference and explain how he would overcome this problem.

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(b) Describe how codecs would be used in the transmission of data within a video-conference.

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9 There are two types of encryption, symmetric and asymmetric.

(a) Describe the use of public keys and private keys when used in the asymmetric method of encryption.

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(b) Apart from the types of keys used, give **two** reasons why symmetric encryption is preferred in certain situations.

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10 Lastri uses spreadsheet models for her company. She uses them to forecast trends of sales of particular products as well as forecasting the company's annual income, expenditure and profits.

By weighing up the advantages and disadvantages, evaluate the effectiveness of spreadsheet models when used specifically for this type of forecasting.

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11 Describe what is meant by normalisation. Include in your description the properties of first normal form (1NF) and how a table in first normal form can be converted to second normal form (2NF).

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12 Sujatmi runs a company which stores its payroll records and personal details of its workers in two flat files. She has been told that these files should be converted into a relational database.

By weighing up the advantages and disadvantages, evaluate the differences between a flat file database and a relational database which would help Sujatmi make her decision as to which to use.

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13 Pramana has a PC which has a fixed hard disk drive. In order to increase the storage capacity of the computer, he is trying to choose between using a portable hard disk drive and storing his data in the cloud.

By weighing up the advantages and disadvantages, evaluate the use of cloud storage compared to using a portable hard disk drive.

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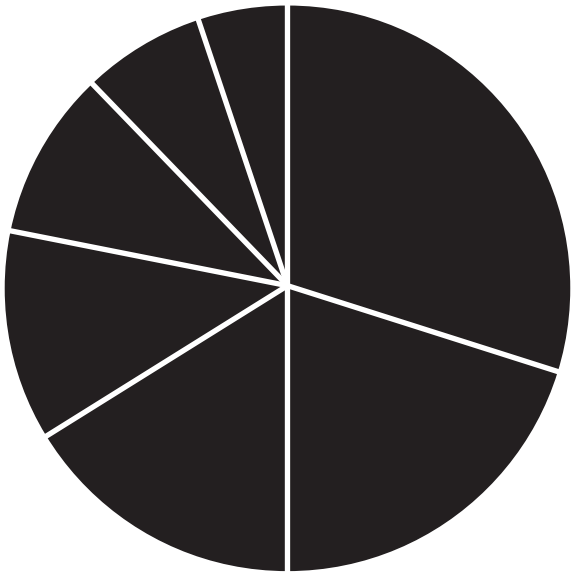
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14



The pie chart shows the land areas of seven continents. In its present form it is not providing any information.

(a) Describe, using examples related to this scenario, how you would improve this chart so that it will provide information which could be used by geography students in a school.

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Please turn over for Question 14(b).

(b) Some people are very critical about pie charts.

Describe the disadvantages of using pie charts to represent data.

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6777728387 4

INFORMATION TECHNOLOGY **9626/11**

Paper 1 Theory **May/June 2021**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 90.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Any blank pages are indicated.

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[Turn over

1 **(a)** Data can exist in two states, dynamic or static.

Tick the most accurate statement referring to dynamic data.

	✓
A sentence in a textbook that says, 'The American Declaration of Independence was in 1776' is an example of dynamic data	
Data that is read from and not written back to a file is called dynamic data	
A hard copy newspaper is an example of a dynamic data source	
Dynamic data can be updated quickly	

[1]

(b) Tick the scenario that involves the use of an indirect data source.

	✓
Handing out a questionnaire to factory workers to find out about working conditions	
Using sensors to monitor pollution in a river	
Asking businesses for their customers' details to produce a mailing list	
Observing the number of customers entering a store to see how busy it is	

[1]

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When data is entered into a database, verification is often used.

(c) Analyse, including their effectiveness, the different methods of verification that can be used when entering data into a database.

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2 There are a number of different ways that data can be directly input to a computer.

(a) Tick the most accurate statement referring to the inputting of text using an Optical Character Reader.

	✓
It is a lot slower to enter large amounts of text than an inexperienced user typing it in	
It can be used to read ordinary handwriting	
It is always completely accurate, with no mistakes being made	
It is more expensive to scan the text than to pay someone to type in the text	

[1]

(b) Tick the most accurate statement referring to the inputting of examination paper answers using an Optical Mark Reader.

	✓
It is very suitable for entering text	
It can read marks made on an otherwise blank piece of paper	
It is much more accurate than data being keyed in by a person	
If the marks are not in a dark enough pencil, they are still always read correctly	

[1]

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(c) Evaluate, by weighing up the advantages and disadvantages, the use of MICR for inputting text.

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3 Financial reports can be produced using word processing software or spreadsheet software.

Compare and contrast the use of these two pieces of application software for this purpose.

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4 Describe how the data from an air pollution monitoring system is collected and prepared, ready for processing by a computer.

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5 A school has classrooms where students can use computers to access the school network. There are other classrooms where the computers do not have access to the school network.

Compare and contrast the use of computers in both types of classroom.

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..... [6]

6 The World Wide Web is a service provided by the internet.

Describe, without referring to how it is accessed, what is meant by the World Wide Web.

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7 The manager of an IT department has been asked to set up a video-conference. At the moment she only has a router, a networked computer and a large screen.

Describe how she would set up this video-conference before joining in.

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8 A company sells domestic appliances. The spreadsheet shows the sales of washing machines for the first five months of this year.

	A	B	C	D	E	F	G
1	**Month**	**Serial Number**		**Price**			
2	Jan	614481		$481		**Sales Month**	**Monthly Total**
3	Jan	707617		$617		Jan	$1,964
4	Jan	227616		$616		Feb	$1,728
5	Jan	907250		$250		Mar	$392
6	Feb	485231		$231		Apr	$815
7	Feb	318299		$299		May	$1,100
8	Feb	211500		$500			
9	Feb	052066		$66			
10	Feb	376632		$632			
11	Mar	086392		$392			
12	Apr	392059		$59			
13	Apr	757756		$756			
14	May	284074		$74			
15	May	517520		$520			
16	May	336250		$250			
17	May	134256		$256			
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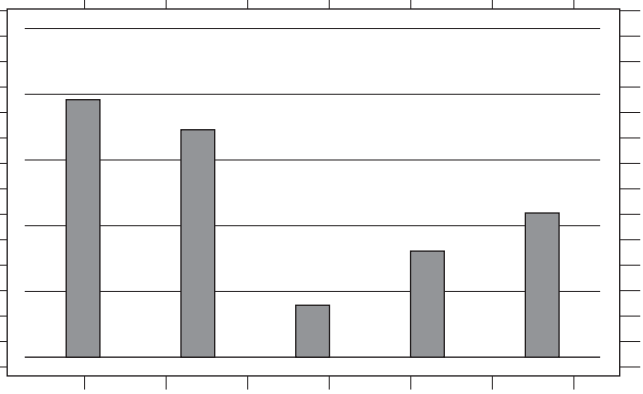


Fig. 8.1

(a) The formula in cell D2 is =VALUE(RIGHT(B2,3)).

Explain, without using cell references, what this formula does.

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(b) In Fig. 8.1, the formula in cell G3 is =SUMIF(A$2:A$17,F3,D$2:D$17).

Explain, without using cell references, what this formula does.

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(c) The bar chart in Fig. 8.1 lacks information about what it represents. It has been created from contiguous data.

(i) Identify the cell range that was used to create the bar chart.

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..... [1]

(ii) Describe what information should be added to the bar chart.

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(d) Evaluate, by weighing up the advantages and disadvantages, the use of the spreadsheet in Fig. 8.1 for financial forecasting.

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9 Metropolis City Library stores details of its books and borrowers in a database. Part of the Borrowers table is shown in Fig. 9.1.

BorrowerID	FamilyName	FirstName	Phone	Town_lived_in
1	Gale	Clemantine	01164960567	Mintair
2	Brown	Christine	01134960965	Redleigh
3	Leadbetter	Karla	07700900863	Lowtown
4	Brierley	Fred	01134960675	Redleigh
5	Greenhalgh	Ben	01164960001	Mintair
6	Pooley	Christopher	03069990694	Armley
7	Chalmers	Rio	07700900900	Fromman
8	Lewis	Ruksana	08081570945	Lowtown
9	Patel	Angela	09098790142	Fromman
10	Bopal	Udoka	01614960836	Lowtown
11	Akula	Billy	01914980487	Armley
12	Benson	Wesley	01845632545	Lowtown
13	Breen	Wally	01915647379	Armley

Fig. 9.1

The librarian has created the following query:

Field:	BorrowerID	FamilyName	FirstName	Phone	Town_lived_in
Table:	Borrowers	Borrowers	Borrowers	Borrowers	Borrowers
Sort:					
Show:	☐	☑	☑	☐	☑
Criteria:		Like "B*"			<>"Lowtown"
or:					

Fig. 9.2

(a) (i) Explain what information the librarian is looking for when using the query in Fig. 9.2.

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..... [4]

(ii) Identify the information that the query would output using the data in Fig. 9.1.

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Part of the Books table is shown in Fig. 9.3.

ISBN	Title	Author	Publisher	NumberofCopies	Replacement Cost
9780577185273	Ruinous Orange	Jonathon Roberts	Ellipsoid	3	$16.99
9780593039427	That Was Then	Jaswant Sandhu	Round Moon Publishers	4	$10.00
9780593065907	The Cheap Strangers	Reeta Khan	Round Moon Publishers	1	$18.99
9780752856006	Too Open Too Short	Paul Stevenson	Douglas Books	2	$16.99
9781356892467	The Divide of Decay	Donal Haroldson	Travellor Books	2	$19.99
9781409187276	Light and Day	Gabrial Aston	Betelgeuse	2	$18.99
9781544196873	Never the Poor	Jaswant Sandhu	Round Moon Publishers	3	$18.99
9781772460886	Gloom Brother	Rowena Littleton	Douglas Books	1	$20.99
9781772469285	Tempest Brother	Rowena Littleton	Douglas Books	1	$13.99
9781780835991	Ambuscade	Hortense de Vandre	Possum Books	2	$12.99
9781781254293	Significant Numbers	Jan Tudor	Frowpile Books	4	$17.99
9781846962345	Respeto por la Castilla	Celia Rodriguez	Possum Books	4	$9.99

Fig. 9.3

The librarian has created the following query:

Field:	ISBN	Title	Author	Publisher	NumberofCopies	ReplacementCost
Table:	Books	Books	Books	Books	Books	Books
Sort:						
Show:	☐	☑	☑	☐	☑	☐
Criteria:			Like "*son"			
or:						< 10

Fig. 9.4

(b) (i) Explain what information the librarian is looking for when using the query in Fig. 9.4.

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..... [2]

(ii) Identify the information that the query would output using the data in Fig. 9.3.

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10 A company is creating a database to record information about each employee. The IT manager is creating an on-screen data entry form that will be used by the Human Resources department. This is her first design:

Name___

Address_____

Date of birth_____

Gender_____

Date joined company_____

Salary_____

Describe, in detail, appropriate improvements that could be made to the form to make data entry more accurate and easier to complete.

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11 Explain, using examples, what is meant by the term generic file format.

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12 Describe, in detail, what is meant by a hierarchical database management system.

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INFORMATION TECHNOLOGY **9626/12**

Paper 1 Theory **May/June 2021**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 90.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Any blank pages are indicated.

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[Turn over

1 **(a)** Tick the most accurate statement referring to the use of a length check when validating data.

	✓
It returns an error message if a data item is not in a lookup list.	
It checks to make sure a data item is within an upper limit and a lower limit.	
It checks that a data item consists of a set number of characters, not more or less.	
It is only used to check numbers.	

[1]

(b) Tick the most accurate statement referring to the use of a check digit when validating data.

	✓
It is used to check that the data in two fields correspond.	
It checks the number of digits.	
It is the digit added to the end of a long number after a calculation has been performed.	
It checks that a number is above a certain limit.	

[1]

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(c) The table shows data relating to a company's products. The data can only take the values shown.

Product number	Weight (kg)	Year patented
QY648	8.2	2020
EA721	9.0	2019
HH419	9.8	2020
CG297	8.6	2019
HE456	9.0	2018
GC684	8.2	2020
KZ638	9.8	2016

Analyse the effectiveness of the use of type checks and format checks to validate this set of data. Do **not** refer to length checks or check digits in your answer.

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2 Sensors are used in both monitoring and control applications.

(a) Tick the most accurate statement referring to the use of technology to monitor weather conditions.

	✓
Readings from sensors are not processed before being output.	
A sound sensor is the sensor most frequently used in a weather station.	
Rainfall is measured using a computerised measuring stick.	
Graphs can be produced automatically from a weather monitoring system.	

[1]

(b) Tick the most accurate statement referring to the use of technology when used in systems which control traffic lights.

	✓
An actuator is the most important input device in such a system.	
The timing sequence of the traffic lights is randomly generated.	
An induction loop is used to detect the presence of an automobile.	
Traffic light systems use moisture sensors to detect the presence of pedestrians.	

[1]

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(c) Describe the inputs, processing and outputs involved in the cooking of a meal using a microprocessor-controlled oven. Include the devices required in your answer.

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3 A manager wants to code the data about clothes sold in a shop, ready for storing in a database. The table shows the manager's first attempt.

Category	**Data**
Colour	Light green (G), Black (B), Dark green (G), Red (R)
Type of clothing	Shirt (S), Skirt (S), Trousers (T), Pullover (P)
Size (ins)	25 – 30 (S), 31 – 36 (L), 37 – 42 (XL), 43 – 48 (XXL)
Brand	Astio (A), Dasiap (D), Utap (U), Gumho (G)

(a) Give **three** different reasons why the manager wants the data coded.

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[3]

(b) Give **three** different drawbacks of coding the data as shown in the table.

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[3]

(c) Describe **two** different improvements which could be made to this coding system.

1 .....

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2 .....

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[2]

4 Define the following types of malware, including a description of what each type does.

(a) Worm

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(b) Spyware

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5 Analyse the effectiveness of using a line graph, compared to a vertical bar chart, for showing shopping trends, using the following data.

2020												
Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Sales ($)	1050	1120	1180	1150	1140	1160	1170	1190	1180	1200	1190	1220

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6 Narinder works as a computer operator for a company which has its own LAN.

(a) Describe the characteristics of a LAN, including the technologies used.

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(b) The company Narinder works for is in Ghana. Her parents live in Tanzania, a long way from Ghana.

She writes a blog and uses instant messaging to stay in touch with her parents.

Describe **four** other methods by which Narinder could use the internet to communicate with her parents, giving a different drawback of each method.

Method 1 .....

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Drawback .....

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Method 2 .....

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Drawback .....

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Method 3 .....

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Drawback .....

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Method 4 .....

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Drawback .....

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[8]

7 Rajvinder is the regional manager of a team of travelling salespeople. Every month he organises a web-conference for his team.

The necessary equipment and software have already been purchased and tested.

Describe how Rajvinder would set up this web-conference and how his team would join the conference.

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8 A common feature of many database systems is the use of switchboards.

Evaluate, by weighing up the advantages and disadvantages, the use of switchboards in such systems.

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9 GoodHope Ltd uses computers to process its payroll. The company pays its workers weekly, which involves the updating of a master file.

Part of a transaction file is shown. It contains workers' ID numbers and the hours worked by those workers in a particular week.

Part of the master file used by the company is shown. It contains the workers' ID numbers and the hourly rate at which they are paid in $.

Transaction file

ID_number	Hours_worked
25306	35
25376	40
25891	37
27813	40
28197	35
32678	40
33429	37
36802	38
38439	38
39286	40

Master file

ID_number	Hourly_rate
13487	26
25306	25
25376	25
25891	24
26145	25
27813	22
28197	23
31026	22
32678	26
33429	22
36802	21
38439	25
39286	24
39875	25

(a) The transaction file has been sorted so that the updating can begin.

Describe what other process must happen before updating can begin.

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..... [1]

(b) Describe the steps involved in updating the master file in this payroll system.
You must use examples from the files shown.
You may assume that the only transaction being carried out is the calculation of the weekly pay using the formula:

Hourly rate × Hours worked

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10 Many nuclear reactions are simulated rather than being carried out in real life.

(a) Describe **three** advantages of using a computer-based model to run a simulation for these reactions.

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(b) Describe **two** disadvantages of using a computer-based model to run a simulation for these reactions.

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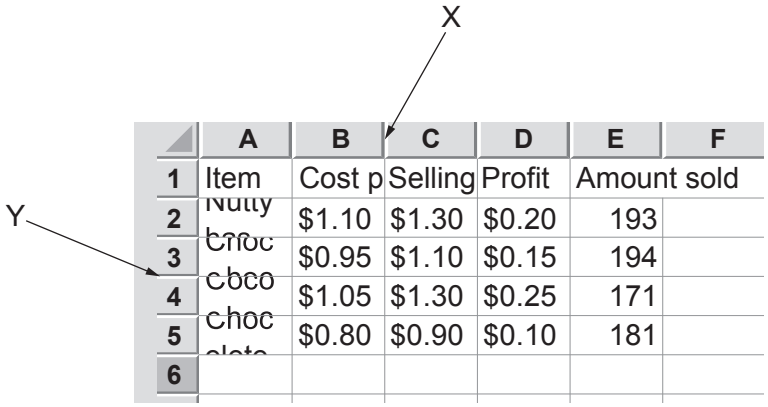
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..... [2]

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11 The spreadsheet below shows information about sweets that a shop sells. The data has been formatted but the spreadsheet needs some improvements.



(a) Describe, in detail, what would happen within the spreadsheet if the mouse was double-clicked with the pointer positioned at X.

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..... [1]

(b) Describe, in detail, what would happen within the spreadsheet if the mouse was double-clicked with the pointer positioned at Y.

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..... [1]

(c) Originally, the data was entered into the cells in the range B2 : C5 as numbers.

Describe the formatting that has been applied to the cells in this range.

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..... [1]

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(d) Validation is to be applied to cells B2 to B5.
This will prevent numbers less than the current minimum value from being entered.
The shop owner will not use a range check or a lookup check.

Name and describe the validation check that will be used.

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..... [2]

Please turn over for Question 12.

12 Joanne has created a spreadsheet using proprietary software. Paula has suggested that she should save it in an open-source file format.

(a) Describe **three** distinct features of proprietary software.

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(b) Describe **two** distinct features of an open-source file format.

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INFORMATION TECHNOLOGY **9626/13**

Paper 1 Theory **May/June 2021**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
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- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 90.
- The number of marks for each question or part question is shown in brackets [ ].

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1 **(a)** A researcher has to choose between using a direct data source or an indirect data source. The data collected may be confidential and may need to be encrypted when sent over the internet.

Tick the statement that best describes the use of a direct data source.

	✓
Searching the World Wide Web to find weather data	
Interviewing residents living on a street to find out how busy the traffic on the street is	
Examining the census results for previous years	
Looking at a list of names and addresses of everyone who is registered to vote	

[1]

(b) Data can exist in two states, dynamic or static.

Tick the most accurate statement referring to dynamic data.

	✓
Dynamic data is data that never changes.	
Data on a hard disk is an example of dynamic data.	
‘Canada has a border with the United States’ is an example of dynamic data.	
Dynamic data is never checked for accuracy before publication.	

[1]

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(c) The SSL/TLS protocol is an encryption protocol frequently used when data is transmitted over the internet.

Describe how the SSL/TLS protocol is used to ensure the security of the data.

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2 Many computer users use a hard disk drive to store their data and use the cloud to back up this data.

(a) Tick the most accurate statement referring to the cloud.

	✓
It is a cheaper method than any other form of storage.	
Data is more secure than when stored using any other form of storage.	
Users have an unlimited amount of free storage.	
You can store data wherever you are, as long as there is an internet connection.	

[1]

(b) Tick the most accurate statement referring to a hard disk drive.

	✓
It is an optical method of storage.	
It can be used to store operating systems.	
It cannot store movies.	
It does not consist of moving parts.	

[1]

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(c) Evaluate, by weighing up the advantages and disadvantages, the use of a magnetic tape drive compared to the use of an external solid-state drive (SSD) for making backups.

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3 Explain how the following tasks are carried out using utility software:

(a) data compression

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(b) disk formatting

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(c) file deletion.

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4 Many supermarkets employ checkout operators. Their job is to sit at a terminal for a set number of hours, scanning products bought by customers and handling payments. Information Technology (IT) is used to monitor the performance of these checkout operators.

Evaluate, by weighing up the advantages and disadvantages to the employer, the use of IT for this purpose.

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5 A company has several IT rooms. The computers in each IT room are connected to the company's network. One of the physical safety issues is workers tripping over cables.

Describe, in detail, **three** other physical safety issues that can arise from the use of IT rooms. For each issue, give one method of helping to prevent it. Each method must be different.

Issue 1 .....

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Prevention method .....

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Issue 2 .....

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Prevention method .....

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Issue 3 .....

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Prevention method .....

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..... [6]

6 A company has to decide between using a peer-to-peer or a client-server computer network.

(a) Describe the differences between client-server and peer-to-peer computer networks.

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The company will use its network to access the internet.

(b) Describe **three** benefits to companies of using the internet to conduct business.

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(c) Describe **four** drawbacks to companies of using the internet to conduct business.

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7 Explain what is meant by the following technologies when used in video conferencing:

(a) ISDN

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(b) SDSL

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8 The spreadsheet shows the values of used cars and the deposits required to buy them.

D2 | =ROUND(C2*10%,0)

	A	B	C	D
1	Car make	Car model	Car value	Deposit required
2	Diap	D5	$4,752.00	$475.00
3	Diap	D1	$6,732.00	$673.00
4	Sentel	M810	$6,655.00	$666.00
5	Sentel	R500	$5,445.00	$545.00
6	Panther	ZE	$8,470.00	$847.00
7	WRM	WB	$10,285.00	$1,029.00
8	WRM	WA	$12,100.00	$1,210.00
9	Diap	D4	$4,235.00	$424.00
10	Panther	ZK	$9,438.00	$944.00
11	WRM	WB	$7,502.00	$750.00
12	Diap	D3	$5,082.00	$508.00
13	Panther	ZJ	$10,890.00	$1,089.00
14	Sentel	G47	$11,374.00	$1,137.00
15	WRM	WB	$10,527.00	$1,053.00

(a) Explain what is meant by the term ‘formula’. Refer to the contents of cell D2 in your answer.

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(b) Explain what is meant by the term ‘function’. Refer to the contents of cell D2 in your answer.

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..... [3]

The spreadsheet has now been sorted.

	A	B	C	D
1	Car make	Car model	Car value	Deposit required
2	Diap	D5	$4,752.00	$475.00
3	Diap	D4	$4,235.00	$424.00
4	Diap	D3	$5,082.00	$508.00
5	Diap	D1	$6,732.00	$673.00
6	Panther	ZK	$9,438.00	$944.00
7	Panther	ZJ	$10,890.00	$1,089.00
8	Panther	ZE	$8,470.00	$847.00
9	Sentel	R500	$5,445.00	$545.00
10	Sentel	M810	$6,655.00	$666.00
11	Sentel	G47	$11,374.00	$1,137.00
12	WRM	WB	$7,502.00	$750.00
13	WRM	WB	$10,285.00	$1,029.00
14	WRM	WB	$10,527.00	$1,053.00
15	WRM	WA	$12,100.00	$1,210.00

(c) Identify the steps that have been taken by a user to produce this sorted spreadsheet.

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9 Johan has produced a spreadsheet.

Describe **five** advantages of Johan saving the spreadsheet as a csv file compared to saving it as a spreadsheet file.

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10 A company employs over one thousand workers in a multi-storey office block. Sometimes fire drills are carried out and workers have to leave the building. The company wishes to know if the use of a computer simulation would be a better method of discovering how easy it would be to empty the building.

Evaluate, by weighing up the advantages and disadvantages, the use of a computer simulation for this scenario.

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11 Explain what is meant by normalisation. Include in your answer a description of the characteristics of second normal form (2NF) and third normal form (3NF).

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12 A management information system (MIS) can be used to produce reports.

Describe other features of a management information system (MIS).

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..... [5]

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INFORMATION TECHNOLOGY **9626/11**

Paper 1 Theory **May/June 2022**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
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INFORMATION

- The total mark for this paper is 70.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages. Any blank pages are indicated.

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1 The processing of customer orders is an example of batch processing.

Describe, using customer orders as an example, what is meant by batch processing.

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2 Flight simulators are used to train pilots to fly modern jet aircraft.

Describe a typical flight simulator.

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3 When personal data is sent across the internet, it is often encrypted.

Analyse the encryption of personal data when it is sent across the internet.

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4 A supermarket uses a computer to automatically update stock levels. Each record in the **Product file** contains the fields shown.

Product file

Bar_code	Product	Number_in_stock	Reorder_level	Reorder_quantity	Supplier_ID
5022476130067	Oaters	131	130	250	B671841
0100108097858	Tea	182	130	310	A296593
0201570244453	Beans	317	212	450	C562148

When the Number_in_stock of a product item falls to the Reorder_level, that product must be reordered. The Reorder_quantity is the number of items of the product that must be reordered.

The **Supplier file** contains the contact details of all the suppliers.

Supplier file

Supplier_ID	Supplier	Contact
B671841	J Sullivan	js@cmail.co.ke
A296593	Bluants	bluants@hmail.ug
C562148	Golan	golan@smail.eg

Only some of the records in the Product and Supplier files are shown.

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Describe, using the Oaters record as an example, the steps the computer would follow to update the Product file after a point of sale terminal reads the bar code for a packet of Oaters.

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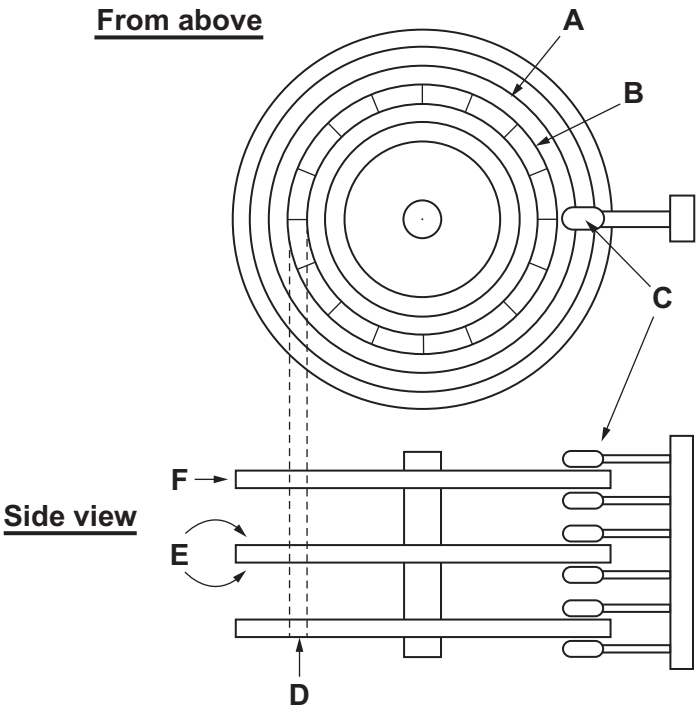
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5 Here is a diagram showing the component parts of a hard disk drive.



(a) Describe the component parts of a hard disk drive. Refer to the labelled diagram in your answer.

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(b) Explain what is meant by disk defragmentation and why it is needed when using hard disks.

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..... [4]

6 Many houses in cooler countries have central heating systems. In a central heating system, microprocessors are used to control the pump which sends water from the boiler to the individual heaters.

Complete the pseudocode algorithm to show the processing which takes place in this microprocessor-controlled central heating system. The algorithm must prevent attempting to switch the pump on when it is already on and attempting to switch the pump off when it is already off. You may assume `temperature` is the variable representing the actual temperature of the house and `preset` represents the required temperature of the house.

```
REPEAT
INPUT temperature
    IF temperature < preset
        THEN
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..... [6]

7 Automatic re-calculation is one of the characteristics of spreadsheet models. This means it is not necessary to evaluate a formula every time you change it.

Describe other characteristics of spreadsheets that make them suitable for creating models.

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8 Ruritania has just passed a law which will allow supermarkets to purchase customer information from banks, providing they anonymise it. Joanne, the owner of a supermarket, wishes to analyse the spending habits of people in the area very near to the supermarket. She can either use direct data sources or obtain the information indirectly from local banks.

Discuss the advantages and disadvantages to Joanne of obtaining the information indirectly from local banks.

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9 One reason for the digital divide is that people living in rural areas are less likely to have high speed broadband compared to people in cities.

Explain why this particular aspect of the digital divide exists.

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10 A farmer is concerned that the chemicals he is using to improve his crops are seeping into the local river. He has asked a local conservation group to investigate if this is the case. They will be using a computer and one set of sensors to monitor the level of pollution.

List examples of sensors the group will use and describe how they will be used with the computer to monitor the pollution level.

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11 A gesture based interface is a type of user interface.

Describe the advantages of using a gesture based interface rather than other types of interface.

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INFORMATION TECHNOLOGY **9626/12**

Paper 1 Theory **May/June 2022**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
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INFORMATION

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This document has **12** pages. Any blank pages are indicated.

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1 Some students have been asked by their teacher to carry out a study. They will collect data about the families living in five streets close to their school. They will carry out the study using direct data sources and will not use records from the 2020 census which are kept at the city hall.

Describe the suitability of using direct data sources rather than the census data for the study.

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..... [5]

2 Encryption has many uses, including the encryption of hard disks.

Describe hard disk encryption.

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..... [4]

3 Julia is setting up a hairdressing business. She has recruited several staff. She is creating a database of all her staff and has written down all their details ready for entering into a computer. She will need to verify the data when it is entered.

She has been told that she cannot use a parity check but should use visual verification and double data entry.

(a) By describing what is meant by a parity check, explain why it might not be suitable in this scenario.

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(b) Compare the similarities and differences between the visual checking and double data entry methods of verification.

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4 The performance of mainframe computers and supercomputers is often measured using performance metrics.

Compare, by describing the similarities and differences between them, the different types of performance metrics.

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5 When a high level language program is written, it has to be translated so the computer can run it. This can be achieved using a compiler or an interpreter.

Discuss the advantages and disadvantages of using a compiler rather than an interpreter.

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6 Sensors are used in monitoring systems to measure physical variables. It is important that sensors are calibrated before use.

(a) Describe, in detail, these sensors and give an appropriate use for each. Each use must be different.

Humidity .....

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Sound .....

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[4]

(b) Explain the importance of calibration.

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(c) Describe the differences between one point calibration and two point calibration.

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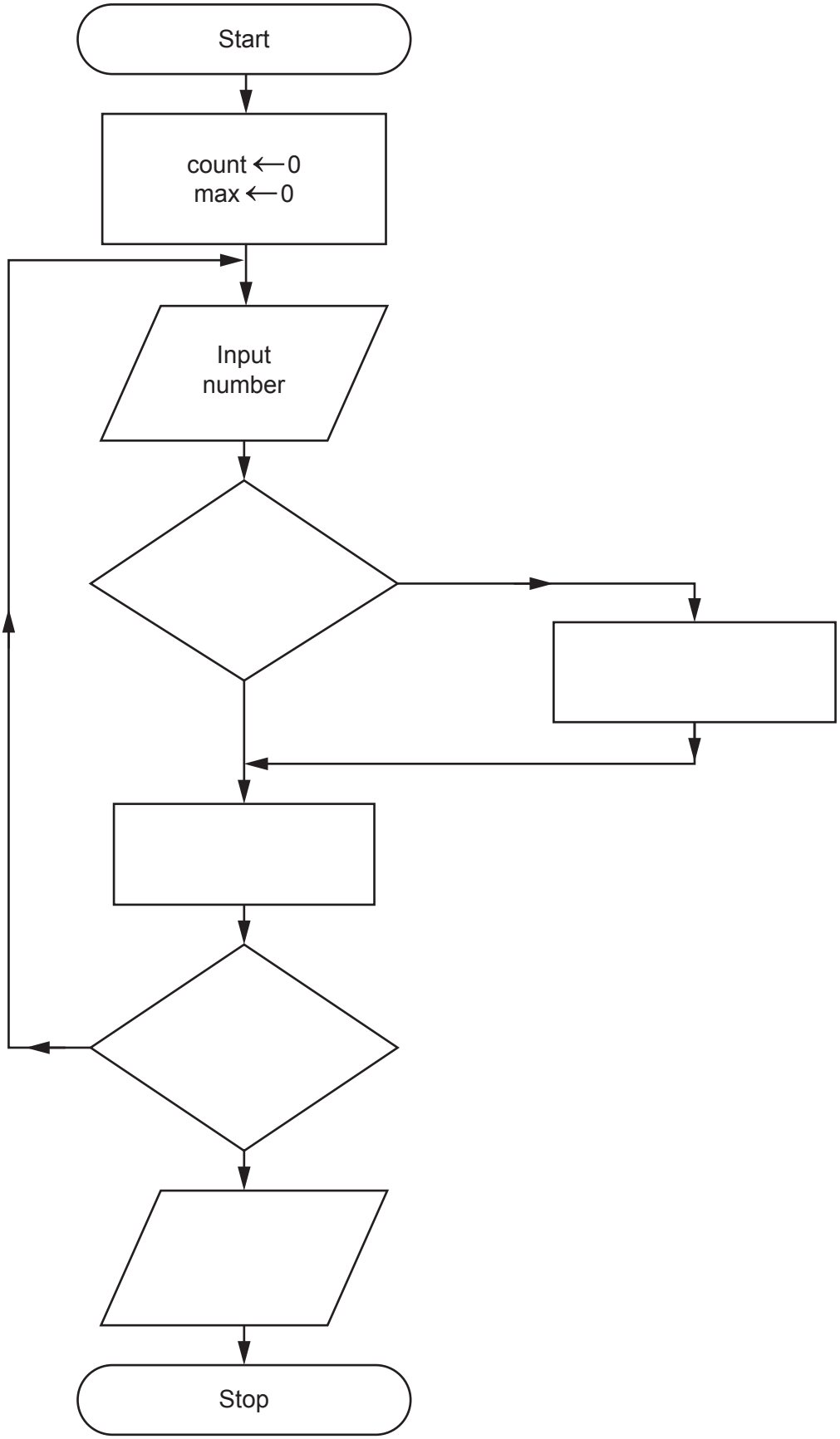
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..... [5]

7 Josefine has started to draw a flowchart which inputs 10 numbers and outputs the largest value.

Complete the flowchart by filling in the empty boxes and by placing yes and no as appropriate in the diagram.



[7]

8 Pharming is a method used by hackers to obtain information from bank customers.

(a) Explain what is meant by pharming.

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(b) Describe **three** methods a customer could use to minimise the risk of becoming a victim of pharming.

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..... [3]

9 Many companies use computers to process their payroll.

A transaction file is produced once a week. It consists of the WorkerID and HoursWorked fields. The master file contains the WorkerID, RateOfPay and WagesSoFar fields as well as other details. The WagesSoFar field represents how much each worker has earned so far this year.

Each worker's wage is calculated by multiplying the HoursWorked by the RateOfPay. When the wage for that week is calculated, it is added on to the WagesSoFar field.

Every week the master file has to be updated to include the new wage earned so far this year.

(a) Complete the following pseudocode algorithm which shows the process of updating the master file. You can assume no other transactions are being carried out.

```
READ first record in transaction file
READ first record in old master file
WHILE not end of transaction file
    IF transaction file WorkerID = master file WorkerID
        THEN
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..... [6]

(b) Updating a master file in a payroll system is an example of batch processing.

Discuss the advantages and disadvantages of using batch processing, compared to other types of processing, for a payroll system.

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INFORMATION TECHNOLOGY **9626/13**

Paper 1 Theory **May/June 2022**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
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INFORMATION

- The total mark for this paper is 70.
- The number of marks for each question or part question is shown in brackets [ ].

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1 Paulo is carrying out a study on the level of noise caused by traffic in the streets around his home. He will need data for his study.

(a) Describe what is meant by direct data.

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..... [1]

(b) Describe how Paulo could use direct data sources for his study.

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2 Explain how high level language programs can be translated to run on different computer systems, including any difficulties which may result from this.

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..... [4]

3 Wei is trying to minimise the effects of malware on her computer system.

Describe the types of software Wei would need and how she could use them to help prevent malware affecting her computer system.

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..... [4]

4 The directors of a company wish to create a database of customers. They are not sure whether they should ask a computer programmer to produce custom written software or they should purchase off-the-shelf software.

(a) Describe the features of custom written software.

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(b) Describe the features of off-the-shelf software.

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5 A doctor drives to work. While at work and while travelling, the doctor uses a number of IT-based devices. She uses a dialogue interface to control these devices.

Discuss the advantages and disadvantages of the doctor using a dialogue interface rather than a graphical user interface (GUI).

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6 Describe how the SSL/TLS protocol is used in client-server networks.

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7 A company has been asked by a government to construct a nuclear reactor. A computer model will be created before the reactor is built.

Explain why a computer model is needed in this scenario.

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8 A microprocessor-controlled car park barrier is used to allow cars to enter a car park. When a car approaches the barrier, the barrier automatically rises to allow the car to enter. It does not lower until the car is safely clear of the barrier system. There is a light sensor and light beam source immediately before the barrier. No tickets are issued by the system. The system operates continuously while it is switched on.

Complete this pseudocode algorithm to describe the processing involved. Your pseudocode algorithm should include IF…THEN statements.

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REPEAT
    INPUT reading from induction loop
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..... [7]

9 Insurance consultants have to make decisions about whether or not to accept a customer's application for an insurance policy. They often use expert systems to help them assess the risk involved.

(a) Such a system would have a knowledge base, which consists of a database of facts and a rules base.

Describe the other components of this expert system.

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(b) Explain how this insurance expert system would be used by the consultant and how it would assess the risk.

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10 Normalisation is used when designing relational databases.

Discuss the advantages and disadvantages of normalisation.

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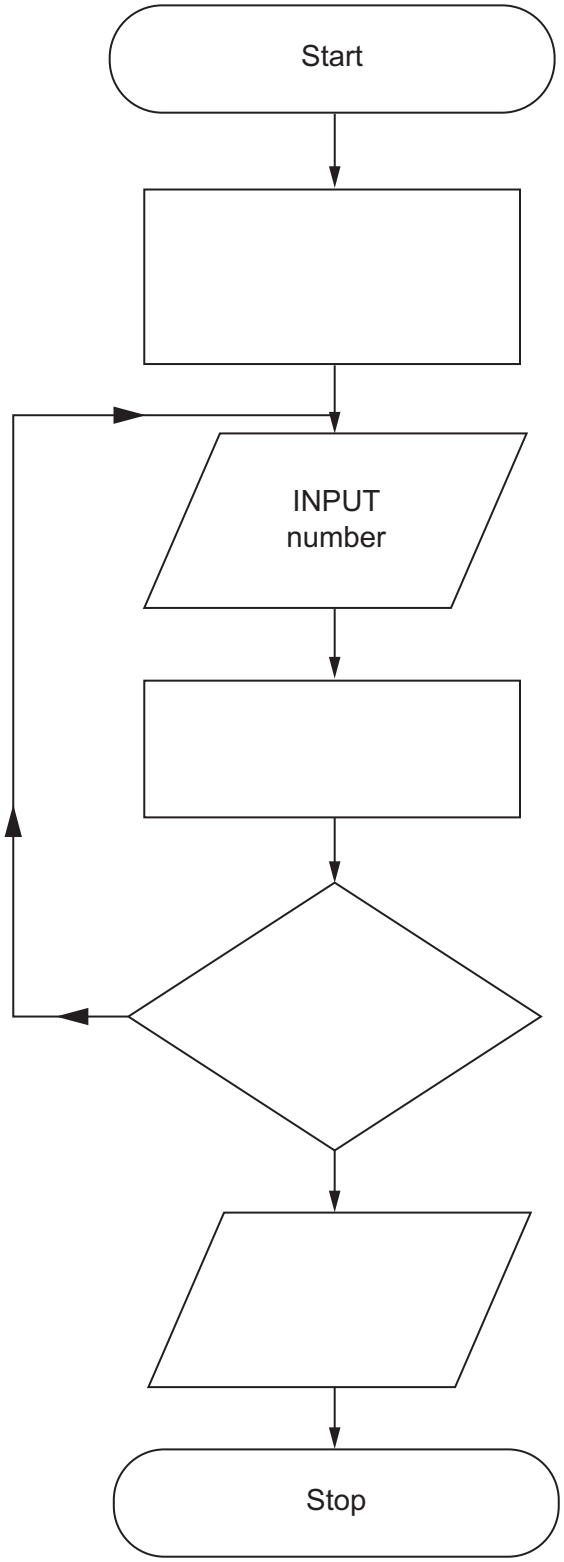
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11 Complete the flowchart below to add up 6 numbers.
Use the variable *total* to store the sum of the numbers and the variable *count* to control the number of times the loop is repeated.



[7]

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INFORMATION TECHNOLOGY **9626/11**

Paper 1 Theory **May/June 2023**

1 hour 45 minutes

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No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
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INFORMATION

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1 Encryption can be used to protect data.

(a) Define encryption.

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(b) Explain how intercepted data could be misused if it was not encrypted.

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2 Describe what is meant by a smart home.

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..... [4]

3 Explain why a compiler may not be suitable for translating a high level language program for use on different computer systems.

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..... [3]

4 The digital divide can exist between people in less industrially developed nations and those in more industrially developed nations.

(a) Explain why this digital divide exists.

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(b) Describe the effects of this form of digital divide.

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5 D'Oliveira Inc. is a company which sells stationery products to many stores. The company uses a transaction file to keep a record of the number of products it has sold each month.

Part of a transaction file is shown. It contains:

- ProductID
- NumberSold (the number of that product sold in a particular month)

Part of the master file used by the company is also shown. It contains:

- ProductID
- Product (description of product)
- NumberinStock (how many of that product are left in stock)

The NumberinStock on the new master file is calculated every month by subtracting the NumberSold from the NumberinStock on the old master file.

Transaction file

ProductID	NumberSold
2210	6500
3356	7300
4268	2900
5690	4600

Master File

ProductID	Product	NumberinStock
1965	Folder wallets	9000
2210	Ring binders	12000
2976	Notepads	10100
3356	A4 paper	13450
4268	A3 paper	8800
5690	HB pencils	9200
6528	Pencil cases	11150
7649	Maths sets	8240

Fig. 5.1

(a) Identify the **two** processes that would have been performed on the transaction file before the master file could be updated.

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..... [1]

(b) List the steps involved when the master file is updated using the transaction file. Use the data shown in Fig. 5.1. Stop at the step where the third master file record is read.

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6 Back-up software is a type of utility software. A network manager wants to back up the data stored on a server.

(a) Describe, with reasons, the steps the manager takes when setting up this process.

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(b) Some of the data to be backed up consist of audio files. The network manager will compress these before starting the back-up process. Different methods of compression can be used.

Describe these methods giving the effects on the audio files of using each method.

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..... [4]

7 A computer system is used to monitor the level of pollution in the air.

Describe how the system would collect, process and display the data.

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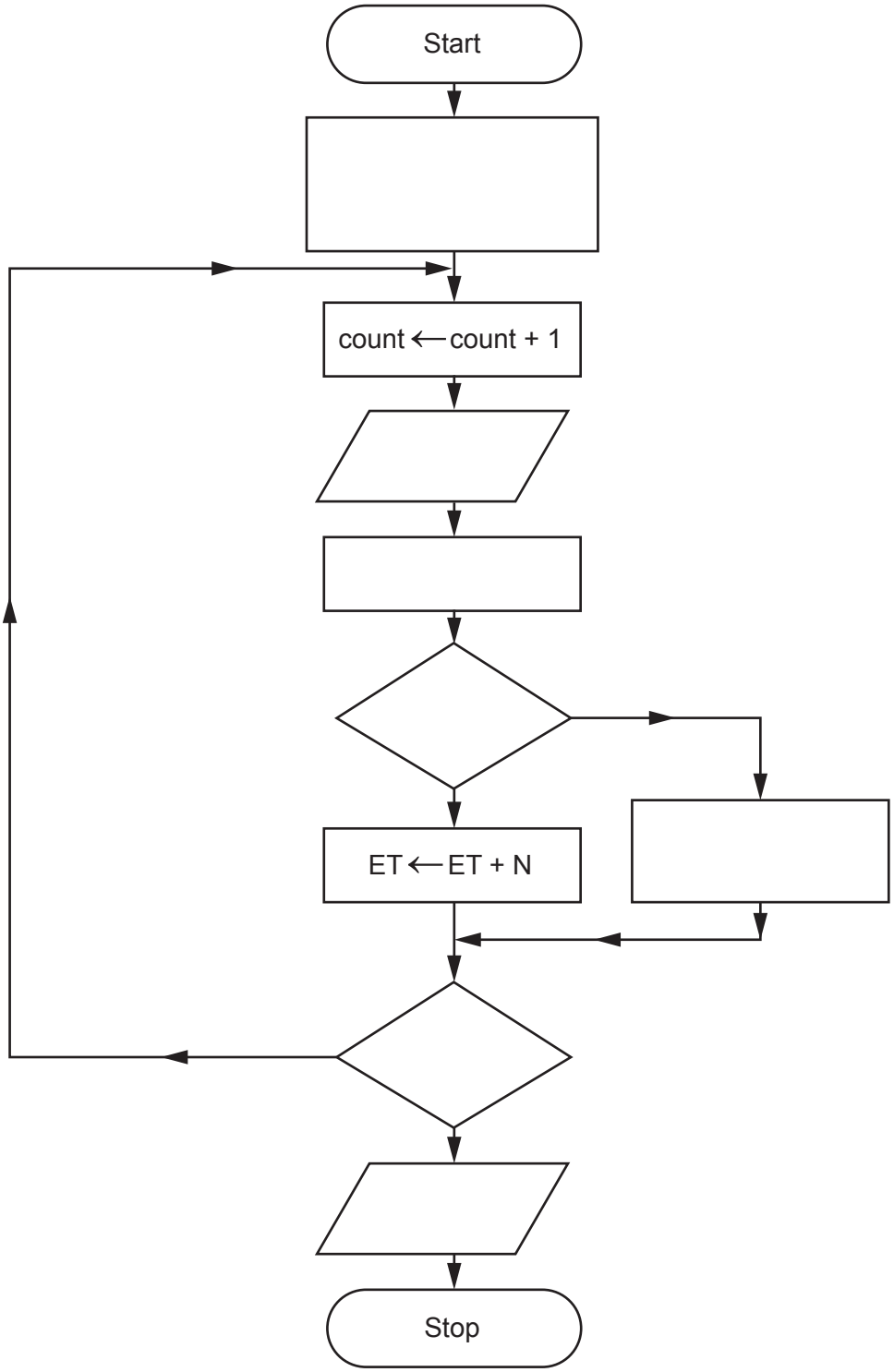
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..... [6]

8 Complete this flowchart which will allow the input of ten numbers.

- A number is even if the number divided by 2 is an integer.
- If a number (N) is odd, it will be added to the odd total (OT).
- If a number (N) is even, it will be added to the even total (ET).
- Both these totals will be output after the tenth number has been processed.



[8]

9 Vishing is a method used by fraudsters to obtain banking information from bank customers by phoning them.

Evaluate the methods that bank customers could use to prevent becoming a victim of vishing.

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10 The quality of sound produced from the data stored on an optical disk depends on the sampling resolution.

Describe what is meant by sampling resolution.

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..... [5]

11 Isabella is concerned about the amount of traffic in her local area. She is going to interview all the nearby families to discover if the amount of traffic is caused mainly by local families driving short distances. It will take her a week to collect the data.

Justify the use of direct sources such as interviews in this scenario.

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INFORMATION TECHNOLOGY **9626/12**

Paper 1 Theory **May/June 2023**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
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INFORMATION

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1 A student has been asked by her sociology teacher to produce a study of the average number of family members in the local area.

(a) Give reasons why using an Electoral Register as an indirect data source would not produce the data required by the study.

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..... [5]

(b) Apart from the Electoral Register, give an example of another indirect data source.

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..... [1]

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2 Explain what is meant by symmetric encryption.

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..... [5]

3 Many credit card purchases use online processing, whereas customer bills tend to be produced using batch processing.

Describe the differences between online processing and batch processing systems.

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4 Some individuals post personal data on social media or in emails.

Discuss why personal data needs to be kept confidential.

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5 A printer driver is a type of device driver.

Explain, using a printer driver as an example, what is meant by a device driver and why it is needed.

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6 Sensors are used in monitoring and control to measure physical variables.

(a) Describe these sensors. For each sensor, explain how it could be used in a control system. Each system must be different.

Infrared .....

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Ultrasonic.....

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[4]

(b) Control technologies are replacing human-controlled car park barriers and timer-controlled traffic lights.

Discuss the advantages and disadvantages of the use of control technologies in these scenarios.

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7 Explain why there can be a digital divide between different socioeconomic groups.

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8 Peter wants to create a database containing the details of all the books he owns and the shops he bought them from. Mary suggests that he should create a relational database. Peter will need to use normalisation.

(a) Describe what is meant by normalisation.

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(b) Describe the benefits of creating a relational database rather than a flat file database.

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9 A doctor uses an expert system to help the doctor diagnose the illnesses of her patients.

Explain how an expert system is used to help the doctor produce the diagnosis of an illness. You may assume only forward chaining is involved.

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10 Gustav has started to write an algorithm using pseudocode. It will input 6 numbers, all less than 1000, and output the smallest value.

Complete the algorithm using pseudocode.

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count ←0
smallest←1000
WHILE count < 6
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..... [6]

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CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

2600220212

INFORMATION TECHNOLOGY **9626/13**

Paper 1 Theory **May/June 2023**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 70.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **12** pages. Any blank pages are indicated.

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1 Describe what is involved in data logging.

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2 Explain the purpose of the SSL/TLS protocol.

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3 There are different types of processing including batch processing and real-time processing.

(a) Without referring to master files and transaction files, describe what is meant by batch processing.

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(b) Rocket guidance systems are used in space exploration.

Explain why a rocket guidance system is an example of the use of real-time processing.

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4 One of the functions of an operating system is to maintain the security of a computer system.

Describe the **other** functions of an operating system.

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5 Justify the use of Information Technology for monitoring the environment.

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6 A computer-controlled traffic light system controls the flow of motor vehicles and pedestrians.

Describe the inputs, processing and outputs involved in this system.

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7 Explain the reasons for the digital divide between the old and the young.

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8 Spyware is a type of malware.

Describe the consequences to the individual of the use of spyware on their computer.

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9 Discuss the advantages and disadvantages of using a gesture-based user interface rather than a graphical user interface to control devices.

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10 An expert system is being trialled for use as part of a study on animal identification. It has been set the following task.

The goal is to decide the colour of a bird named Hortense.
Two facts are provided:

- she sings
- she is 15 cm tall.

The rule base contains the following three rules:

- IF X is an ostrich – THEN X has a long neck
- IF X sings and X is 15 cm tall – THEN X is a canary
- IF X is a canary – THEN X is yellow

The inference engine has concluded that Hortense is yellow. It has taken the facts as data and used the rules base to decide:

1 Hortense is a canary
2 Hortense is yellow

(a) Identify which type of chaining the inference engine used.

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..... [1]

(b) Explain the reasoning the inference engine used.

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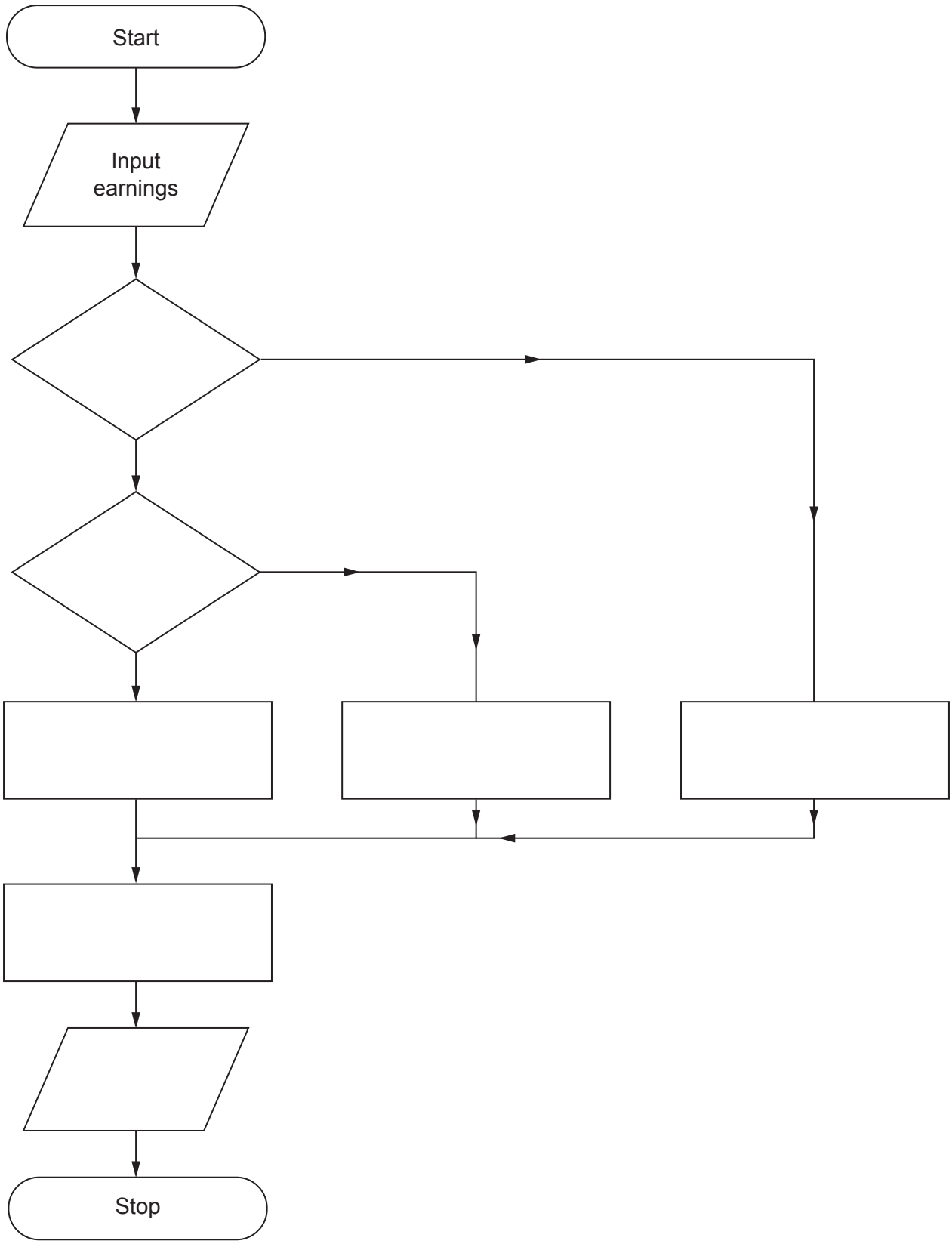
..... [4]

11 In Ruritania these are the tax rules.

- People who earn up to $10 000 pay no tax.
- People who earn between $10 000 and $40 000 pay 20% tax on their earnings **above** $10 000.
- People who earn more than $40 000:
  pay $6 000 ($30 000 × 20%)
  and pay 40% tax on their earnings **above** $40 000.

Complete this flowchart which shows how the net wage (the money earned after tax is paid) is calculated and is output for an individual in Ruritania.

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[8]

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CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

4536664774

INFORMATION TECHNOLOGY **9626/11**

Paper 1 Theory **October/November 2019**

1 hour 45 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your centre number, candidate number and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

Calculators are not allowed on this paper.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Any businesses described in this paper are entirely fictitious.

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1 Tick the **four** most accurate statements regarding the benefits of gathering data from direct data sources.

	✓
You only have to gather as much or as little data as you need	
It allows data to be gathered from subjects to which the data gatherer does not have physical access	
You can obtain the data faster than using an indirect data source	
You have full control over the method used to collect the data	
A much larger sample size can be used than is possible with indirect data sources	
A direct data source is quicker to search	
There may be an opportunity to sell the data to other people for them to use	
You do not have to spend time interviewing people or giving out questionnaires	
A CD ROM of historical data is a direct data source	
The source of the data is known exactly, making it easier to judge its reliability	

[4]

2 Tick the **four** most accurate statements regarding the gathering of personal data by unauthorised persons.

	✓
Pharming involves sending a website URL, inviting the receiver to go to the website	
Smishing involves using text messages to persuade people to give out personal information	
Phishing involves installing malicious software on the victim's computer	
Pharming involves creating a fake website that looks like an actual bank's website	
Phishing involves the sending of fake emails in order to advertise the bank	
Vishing involves the fraudster's computer redirecting a customer's phone call to his own phone	
Banks can ask you from time to time to send them your full password to check you are authorised to use that account	
Vishing involves receiving a phone call from a fraudster	
Vishing never involves getting the customer to phone the bank	
Smishing is the same as vishing but never involves the use of a phone	

[4]

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3 Most computer systems consist of hardware and software.

(a) Define the term 'hardware' in an IT context.

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..... [1]

(b) Giving brief descriptions of **two** examples, define the term utility software.

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4 Using a news website as an example, describe what is meant by dynamic data.

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5 When creating queries, a function IIf() is often used. A table containing a list of where people live and their status could exist so that a query could be constructed:

IIf([Living_in]="own home","own","other")

This would look in a list of where people lived for the words 'own home' and would output the word 'own' if a match were found. If it was not 'own home' then the word 'other' is output.

(a) Complete the nested query that would be used to output 'rented' for anyone living in accommodation owned by a landlord. This is indicated by the word 'landlord' in the *Living_in* field (see table on page 5).

IIf([Living_in]="own home","own" .....

..... [6]

You can use the space below for any working you need.

(b) Below is shown the output created from a larger database.

Describe the steps required to produce this output (from the query in part **(a)**), without using a wizard.

Second_Name	Name1	Living_in	Status
Burke	Dougie	own home	own
Charles	David	with parents	other
Hoy	Tony	landlord	rented
Lee	Duncan	own home	own
Macdonald	Lotte	with friends	other
Norfolk	Jeannie	landlord	rented
Reid	Laura	own home	own
Rydell	Jenna	own home	own
Ryder	Joe	with parents	other
Thomson	Jimmy	landlord	rented

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6 John, a user interface designer, has been asked by a systems analyst to design an interface for a proposed information system. He will need to consider a mental model of the potential users of the system.

(a) Describe what is meant by a mental model in this context.

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(b) Describe the features of a user interface which need to be considered when planning its design. Where appropriate, give examples of each feature.

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7 A hospital has installed video-conferencing equipment so that senior doctors in the hospital can stay in touch with other doctors who work in areas away from the hospital.

Evaluate the impact of video-conferencing on **both** groups of doctors.

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8 Roger has decided to set up a mail order company. He needs to create a database of customer records and orders. He is not sure whether to create a flat file database or a relational database.

Explain the differences between a flat file database and a relational database.

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9 Computer networks can often be classified as consisting of two types, peer-to-peer and client-server networks.

Describe a peer-to-peer network.

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10 Expert systems often use a mixture of forward chaining and backward chaining to determine the probable solution to a problem.

Describe the terms:

(a) Forward chaining.

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(b) Backward chaining.

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11 Here is a spreadsheet showing the medal winners at some of the 2016 Olympics swimming events.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Country code	Country name		Gold	Silver	Bronze		Swimmer	Distance	Stroke	Gender	Position	Code of country		Time (secs)	Average speed (m/s)	Name of country
2																	
3	AUS	Australia		1	0	1		Kyle Chalmers	100m	Freestyle	M	1	AUS		47.58	2.10	Australia
4	BEL	Belgium		0	1	0		Pieter Timmers	100m	Freestyle	M	2	BEL		47.80	2.09	Belgium
5	CAN	Canada		1	1	0		Nathan Adrian	100m	Freestyle	M	3	USA		47.85	2.09	United States
6	CHN	China		1	0	0		Penny Oleksiak	100m	Freestyle	F	1	CAN		52.70	1.90	Canada
7	GBR	Great Britain		1	0	0		Simone Manuel	100m	Freestyle	F	2	USA		52.71	1.90	United States
8	RSA	South Africa		0	2	0		Sarah Sjostrom	100m	Freestyle	F	3	SWE		52.99	1.89	Sweden
9	SWE	Sweden		1	1	1		Adam Peaty	100m	Breaststroke	M	1	GBR		57.13	1.75	Great Britain
10	USA	United States		1	1	4		Cameron Van der Burgh	100m	Breaststroke	M	2	RSA		58.69	1.70	South Africa
11								Codey Miller	100m	Breaststroke	M	3	USA		58.87	1.70	United States
12								Sarah Sjostrom	100m	Butterfly	F	1	SWE		55.48	1.80	Sweden
13								Penny Oleksiak	100m	Butterfly	F	2	CAN		56.46	1.77	Canada
14								Dana Vollmer	100m	Butterfly	F	3	USA		56.63	1.77	United States
15								Yang Sun	200m	Freestyle	M	1	CHN		104.65	1.91	China
16								Chad Le Clos	200m	Freestyle	M	2	RSA		105.20	1.90	South Africa
17								Conor Dwyer	200m	Freestyle	M	3	USA		105.23	1.90	United States
18								Katie Ledecky	200m	Freestyle	F	1	USA		113.73	1.76	United States
19								Sarah Sjostrom	200m	Freestyle	F	2	SWE		114.08	1.75	Sweden
20								Emma McKeon	200m	Freestyle	F	3	AUS		114.92	1.74	Australia

(a) Write down the formula which should go in cell D3 to calculate the number of gold medals that Australia won in the events. The formula should be easily replicable to show the number of gold medals won by the other countries.

The formula should work even if the swimmer details are changed.

= ..... [6]

You can use the space below for any working you need.

(b) The average speed of each swimmer is calculated by dividing the distance swum by the time taken in seconds.

Write down the formula which should go in cell P3 to calculate Kyle Chalmers' speed. The formula should be easily replicable to show the speeds of the other swimmers.

= ..... [5]

You can use the space below for any working you need.

(c) Explain how you would amend the spreadsheet so that the cells between M3 to M20 could only accept the country codes listed. If any other code was input an error message informing the user an error had occurred would be output.

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(d) Explain how you would sort the data so that all the female swimmers were grouped together before the male swimmers and with the furthest distance at the top.

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12 The principal of a college wishes to keep records of all his students' details including exam scores. He is not sure whether he should use a spreadsheet or a database to do this.

Evaluate the use of databases and spreadsheets for such a task.

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CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

6138054074

INFORMATION TECHNOLOGY **9626/12**

Paper 1 Theory **October/November 2019**

1 hour 45 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

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DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

Calculators are not allowed on this paper.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

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1 Tick the **four** most accurate statements regarding gathering data from indirect data sources.

	✓
It may involve having to purchase more data loggers	
Several people would have to be employed to carry out interviews	
It always takes longer to gather data from an indirect data source	
It is impossible to ensure that data was collected from a representative sample	
The data may be out of date	
The sample size is always smaller than if you used a direct data source	
The purpose for which data was collected originally may be quite different from the purpose of the current research	
There may be no data available. The data required has simply never been recorded	
The coding used in the data is never explained	
Compared to direct data sources, using indirect data sources is always very expensive when preparing and carrying out the gathering of data	

[4]

2 Tick the **four** most accurate statements regarding internal hardware components.

	✓
A motherboard has a collection of chips and controllers known as the chipset	
A hard disk drive usually has less storage capacity than the RAM within a computer	
ROM is the memory used to store documents the user is currently working on	
The hard disk drive is used to store application software	
A solid state drive has no moving parts	
RAM is memory that cannot be changed	
A hard disk drive uses flash memory to store data	
RAM stores the start up instructions of a computer even when it is switched off	
A PC can still work without having a CPU	
The arithmetic logic unit and the control unit are part of the CPU in a computer	

[4]

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3 A hockey club secretary is planning to computerise the players' records. For this system she is considering using custom written software.

(a) Describe the drawbacks of using custom written software.

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(b) Describe the benefits of using custom written software.

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4 Doctors often use expert systems to help them make a diagnosis.

Without describing its components explain how an expert system helps the doctor come to a decision regarding the diagnosis.

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5 Here is a spreadsheet showing the top European goal scorers in the 2018 Football World Cup qualification matches.

	A	B	C	D	E	F	G	H	I	J	K
1						Code of country	Footballer	Name of country	Games played	Goals scored	Average
2											
3						POL	Robert Lewandowski	Poland	10	16	
4						SWE	Marcus Berg	Sweden	11	16	
5			Country code	Country name		DEN	Christian Eriksen	Denmark	12	16	
6			POL	Poland		POR	Cristiano Ronaldo	Portugal	9	15	
7			SWE	Sweden		GRE	Kostantinos Mitroglou	Greece	10	12	
8			DEN	Denmark		ITA	Ciro Immobile	Italy	12	12	
9			POR	Portugal		BEL	Romelu Lukaku	Belgium	8	11	
10			GRE	Greece		CRO	Mario Mandžukic	Croatia	11	10	
11			ITA	Italy		POR	André Silva	Portugal	10	9	
12			BEL	Belgium		ITA	Andrea Belotti	Italy	9	8	
13			CRO	Croatia		DEN	Thomas Delaney	Denmark	12	8	
14			SWI	Switzerland		SWI	Haris Seferovic	Switzerland	12	8	
15			ROI	Republic of Ireland		ROI	James McClean	Republic of Ireland	11	8	
16			MON	Montenegro		SWE	Emil Forsberg	Sweden	12	8	
17			NED	Netherlands		MON	Stevan Jovetic	Montenegro	8	7	
18						NED	Arjen Robben	Netherlands	7	6	
19						BEL	Eden Hazard	Belgium	8	6	
20						ROI	Daryl Murphy	Republic of Ireland	8	6	
21											

(a) Write down the formula which should go in cell H3 to display the name of the country for Robert Lewandowski. The formula must be easily replicable.

= ..... [5]

You can use the space below for any working you need.

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(b) Explain, in detail, the steps you would take to amend the spreadsheet so that the players will be put in order of the average number of goals scored per games played for each player. The highest average number should be first in the list.

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6 Many organisations have an intranet and often extend it to form an extranet.

Describe, using examples, what an intranet is and how it could be extended into an extranet.

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7 Sadiq has written a program that he now wishes to convert using a compiler.

(a) There are two types of software. Name them, indicating which type a compiler is.

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(b) Describe the function of a compiler.

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8 Kelvin was given the task of creating a relational database to combine all his company's customer records and sales records (invoices). The tables that were created are shown here.

Customer table

Customer_id	Family_name	First_name	Credit_limit	Loyalty_number
M3764A	Burke	Linda	€1,980	408254117150
N3201B	Charles	Lauren	€2,650	545687952356
N5789A	Floyd	Joe	€1,850	803226234709
L2345A	Hall	John	€2,700	678914325678
Q4012A	Jones	Laura	€3,400	384466646362
L3896C	Li	Kurtis	€2,130	846869163975
M4698B	Norfolk	Kate	€1,250	243375720844
M2589D	Rhodes	Lotte	€3,400	385657814001
M4013B	Sun Wu	Li	€2,240	320031330576
P4352D	Watson	Karla	€2,130	322036694448
P3156C	Williams	Joanne	€1,870	421536345681

Invoices table

Invoice_number	Number_of_items	Item_cost	Customer_ref	Delivered
32178	8	€12.45	P4352D	☑
43189	6	€17.30	N5789A	☑
45237	4	€38.50	M4698B	☐
45964	1	€152.20	L3896C	☑
46579	3	€38.20	N3201B	☐
56931	5	€17.50	M2589D	☑

You can assume that all values shown include the maximum and minimum allowed in each field.

(a) Complete the data dictionary below for the Invoices table. You can assume that all values shown are typical. All field sizes should be appropriate.

Attribute	Data type	Field size	Format	Type of key

[6]

(b) Kelvin wishes to add a calculated field showing the total cost for each customer.

Describe the steps he would need to follow in order to create this calculated field using a query without using a wizard.

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(c) The following data was typed in error. For each item explain how validation could have prevented the error and explain what would be allowed.

(i) 321789 in the Invoice_number field.

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..... [2]

(ii) 12500 in the Credit_limit field.

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..... [2]

(iii) L21789 in the Customer_id field.

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..... [2]

(iv) 679814325678 in the Loyalty_number field for John Hall.

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9 Evaluate the use of HTTPS in websites.

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10 Many organisations make use of Management Information Systems (MIS). Explain what a typical MIS consists of, including the role of an MIS manager.

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11 Juanita, an office manager, has just been given the task of producing customer invoices. Each invoice will need to have three copies. She is trying to decide between using a dot matrix printer or an inkjet printer to produce the invoices.

(a) Give **three** detailed advantages of Juanita choosing a dot matrix printer.

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(b) Give **three** detailed disadvantages of Juanita choosing a dot matrix printer.

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12 Laura, the manager of a new company, has just purchased some computers for workers to use. With the increase in the use of ransomware by hackers she has decided that the workers will not be allowed access to the internet. However, she needs to decide whether or not to connect the computers together to form a network.

Without referring to the internet, discuss the advantages and disadvantages of networking these computers.

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CANDIDATE NAME	
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6108546698

INFORMATION TECHNOLOGY **9626/13**

Paper 1 Theory **October/November 2019**

1 hour 45 minutes

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your centre number, candidate number and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.

Calculators are not allowed on this paper.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Any businesses described in this paper are entirely fictitious.

DC (CE/SG) 168823/2
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1 Tick the **four** most accurate statements regarding the benefits of gathering data from direct data sources.

	✓
You only have to gather as much or as little data as you need	
It allows data to be gathered from subjects to which the data gatherer does not have physical access	
You can obtain the data faster than using an indirect data source	
You have full control over the method used to collect the data	
A much larger sample size can be used than is possible with indirect data sources	
A direct data source is quicker to search	
There may be an opportunity to sell the data to other people for them to use	
You do not have to spend time interviewing people or giving out questionnaires	
A CD ROM of historical data is a direct data source	
The source of the data is known exactly, making it easier to judge its reliability	

[4]

2 Tick the **four** most accurate statements regarding the gathering of personal data by unauthorised persons.

	✓
Pharming involves sending a website URL, inviting the receiver to go to the website	
Smishing involves using text messages to persuade people to give out personal information	
Phishing involves installing malicious software on the victim's computer	
Pharming involves creating a fake website that looks like an actual bank's website	
Phishing involves the sending of fake emails in order to advertise the bank	
Vishing involves the fraudster's computer redirecting a customer's phone call to his own phone	
Banks can ask you from time to time to send them your full password to check you are authorised to use that account	
Vishing involves receiving a phone call from a fraudster	
Vishing never involves getting the customer to phone the bank	
Smishing is the same as vishing but never involves the use of a phone	

[4]

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3 Most computer systems consist of hardware and software.

(a) Define the term 'hardware' in an IT context.

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..... [1]

(b) Giving brief descriptions of **two** examples, define the term utility software.

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4 Using a news website as an example, describe what is meant by dynamic data.

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5 When creating queries, a function IIf() is often used. A table containing a list of where people live and their status could exist so that a query could be constructed:

IIf([Living_in]="own home","own","other")

This would look in a list of where people lived for the words 'own home' and would output the word 'own' if a match were found. If it was not 'own home' then the word 'other' is output.

(a) Complete the nested query that would be used to output 'rented' for anyone living in accommodation owned by a landlord. This is indicated by the word 'landlord' in the *Living_in* field (see table on page 5).

IIf([Living_in]="own home","own" .....

..... [6]

You can use the space below for any working you need.

(b) Below is shown the output created from a larger database.

Describe the steps required to produce this output (from the query in part **(a)**), without using a wizard.

Second_Name	Name1	Living_in	Status
Burke	Dougie	own home	own
Charles	David	with parents	other
Hoy	Tony	landlord	rented
Lee	Duncan	own home	own
Macdonald	Lotte	with friends	other
Norfolk	Jeannie	landlord	rented
Reid	Laura	own home	own
Rydell	Jenna	own home	own
Ryder	Joe	with parents	other
Thomson	Jimmy	landlord	rented

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6 John, a user interface designer, has been asked by a systems analyst to design an interface for a proposed information system. He will need to consider a mental model of the potential users of the system.

(a) Describe what is meant by a mental model in this context.

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(b) Describe the features of a user interface which need to be considered when planning its design. Where appropriate, give examples of each feature.

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7 A hospital has installed video-conferencing equipment so that senior doctors in the hospital can stay in touch with other doctors who work in areas away from the hospital.

Evaluate the impact of video-conferencing on **both** groups of doctors.

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8 Roger has decided to set up a mail order company. He needs to create a database of customer records and orders. He is not sure whether to create a flat file database or a relational database.

Explain the differences between a flat file database and a relational database.

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9 Computer networks can often be classified as consisting of two types, peer-to-peer and client-server networks.

Describe a peer-to-peer network.

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10 Expert systems often use a mixture of forward chaining and backward chaining to determine the probable solution to a problem.

Describe the terms:

(a) Forward chaining.

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..... [3]

(b) Backward chaining.

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11 Here is a spreadsheet showing the medal winners at some of the 2016 Olympics swimming events.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Country code	Country name		Gold	Silver	Bronze		Swimmer	Distance	Stroke	Gender	Position	Code of country		Time (secs)	Average speed (m/s)	Name of country
2																	
3	AUS	Australia		1	0	1		Kyle Chalmers	100m	Freestyle	M	1	AUS		47.58	2.10	Australia
4	BEL	Belgium		0	1	0		Pieter Timmers	100m	Freestyle	M	2	BEL		47.80	2.09	Belgium
5	CAN	Canada		1	1	0		Nathan Adrian	100m	Freestyle	M	3	USA		47.85	2.09	United States
6	CHN	China		1	0	0		Penny Oleksiak	100m	Freestyle	F	1	CAN		52.70	1.90	Canada
7	GBR	Great Britain		1	0	0		Simone Manuel	100m	Freestyle	F	2	USA		52.71	1.90	United States
8	RSA	South Africa		0	2	0		Sarah Sjostrom	100m	Freestyle	F	3	SWE		52.99	1.89	Sweden
9	SWE	Sweden		1	1	1		Adam Peaty	100m	Breaststroke	M	1	GBR		57.13	1.75	Great Britain
10	USA	United States		1	1	4		Cameron Van der Burgh	100m	Breaststroke	M	2	RSA		58.69	1.70	South Africa
11								Codey Miller	100m	Breaststroke	M	3	USA		58.87	1.70	United States
12								Sarah Sjostrom	100m	Butterfly	F	1	SWE		55.48	1.80	Sweden
13								Penny Oleksiak	100m	Butterfly	F	2	CAN		56.46	1.77	Canada
14								Dana Vollmer	100m	Butterfly	F	3	USA		56.63	1.77	United States
15								Yang Sun	200m	Freestyle	M	1	CHN		104.65	1.91	China
16								Chad Le Clos	200m	Freestyle	M	2	RSA		105.20	1.90	South Africa
17								Conor Dwyer	200m	Freestyle	M	3	USA		105.23	1.90	United States
18								Katie Ledecky	200m	Freestyle	F	1	USA		113.73	1.76	United States
19								Sarah Sjostrom	200m	Freestyle	F	2	SWE		114.08	1.75	Sweden
20								Emma McKeon	200m	Freestyle	F	3	AUS		114.92	1.74	Australia

(a) Write down the formula which should go in cell D3 to calculate the number of gold medals that Australia won in the events. The formula should be easily replicable to show the number of gold medals won by the other countries.

The formula should work even if the swimmer details are changed.

= ..... [6]

You can use the space below for any working you need.

(b) The average speed of each swimmer is calculated by dividing the distance swum by the time taken in seconds.

Write down the formula which should go in cell P3 to calculate Kyle Chalmers' speed. The formula should be easily replicable to show the speeds of the other swimmers.

= ..... [5]

You can use the space below for any working you need.

(c) Explain how you would amend the spreadsheet so that the cells between M3 to M20 could only accept the country codes listed. If any other code was input an error message informing the user an error had occurred would be output.

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(d) Explain how you would sort the data so that all the female swimmers were grouped together before the male swimmers and with the furthest distance at the top.

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..... [3]

12 The principal of a college wishes to keep records of all his students' details including exam scores. He is not sure whether he should use a spreadsheet or a database to do this.

Evaluate the use of databases and spreadsheets for such a task.

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8127180276

INFORMATION TECHNOLOGY **9626/11**

Paper 1 Theory **October/November 2020**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 90.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages. Blank pages are indicated.

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[Turn over

1 **(a)** Tick the most accurate statement regarding input devices.

	✓
A touch screen is easier to use than a keyboard for inputting large amounts of text	
Using a keyboard is the most efficient method of moving a file from one folder to another	
A scanner is an input device used for inputting hard copy documents	
A motor is an input device used for inputting physical variables in a control system	

[1]

(b) Tick the most accurate statement regarding output devices.

	✓
A sensor is an output device used as an actuator in a control system	
A dot matrix printer is the quickest printer for outputting large amounts of text	
A webcam is an output device used to display moving images	
An inkjet printer can be used to output high-quality photographs	

[1]

(c) Hard disk drives, magnetic tape drives and solid-state drives (SSD) are often referred to as secondary storage devices whereas RAM and ROM are often referred to as primary storage devices.

Explain why computers need secondary storage devices.

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2 **(a)** Tick the statement which most accurately refers to the results of using utility software.

	✓
Lossless compression reduces the number of bits by increasing redundant data	
Some information is lost in lossless compression	
Lossy compression reduces the number of bits by identifying unnecessary information and removing it	
Disk defragmentation organises the contents of the disk into the largest number of contiguous regions	

[1]

(b) Tick the statement which most accurately refers to the properties of a hard disk.

	✓
Data access is faster on a hard disk with fragmented files than on a defragmented hard disk	
Disk formatting prepares a data storage device for initial use	
The first stage of formatting is high-level formatting followed by low-level formatting	
Disk partitioning makes the data storage device invisible to an operating system	

[1]

(c) A type of utility software enables files to be deleted.

Explain how the delete utility works and how files can be recovered even after they appear to have been deleted.

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3 A city suffers from the lack of fresh air because of its heavy traffic. A local government agency uses sensors to monitor the level of pollution in the air. The computer system monitors a number of physical variables.

(a) Describe how this system collects and processes this data.

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(b) Compare and contrast the use of monitoring systems with the use of control systems.

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4 Li has just discovered his printer is not responding when he sends a document to print.

Describe how the following components of an expert system are used to diagnose faults with the printer.

(a) User interface

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(b) Inference engine

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(c) Knowledge base

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..... [2]

5 Louise, a teacher, has been asked to write a text book and will need to submit a sample chapter to the publishers. Before she submits this chapter, she wants to proofread it.

Describe how she should visually proofread this chapter.

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6 DVDs and blu-ray discs are optical media which store data and can be used by computers.

Evaluate, by weighing up the advantages and disadvantages, the use of blu-ray discs compared to DVDs for storing computer data.

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7 Define the following types of malware including a description of what each type does.

(a) Rootkit

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(b) Malicious bots

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(c) Ransomware

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8 Describe the differences between a compiler and an interpreter.

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9 Describe, in detail, the characteristics of a WAN and a LAN.

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10 Below are tables from a university database of Metallurgy students. Relationships link the tables.

Students

Studentid	Phone	Second_name
A5384	01217421600	Bhatti
A6528	01213689624	Schmidt
B7238	01214016620	Jones
A8946	01214006451	Gonzales
D9476	01214016620	Jones

Courses

Studentid	Subject	Teacher_code
A5384	Geology of metals	SMI
A6528	Electrochemical Metallurgy	LOU
B7238	Geology of metals	RIC
A6528	Metallography	SAN
D9476	Physical Metallurgy	SAN

Teachers

Teacher_code	Subject 1	Subject 2
LOU	Electrochemical Metallurgy	Metallography
RIC	Geology of metals	Crystallography
SAN	Metallography	Physical Metallurgy
SMI	Electrochemical Metallurgy	Geology of metals
TUR	Crystallography	Geology of metals

(a) Define, exactly, each of the following terms giving examples of the only suitable fields for each, using the tables provided.

(i) Primary key

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(ii) Foreign key

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(iii) Compound key

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(b) Explain, indicating the type of relationship formed, how the relationships were created connecting these tables. You can assume the tables were already populated with data.

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(c) Describe, in detail, the two validation checks that could be carried out on the Studentid field which could **not** be carried out on the Second_name field.

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..... [3]

11 Amir has created a student database query which allows users to type in any name they choose and then the query finds everybody in the database with that name.

Annisa has told him that this is a dynamic parameter query.

Evaluate, by weighing up the advantages and disadvantages, dynamic parameter queries compared with static parameter queries.

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Please turn over for Question 12.

12 There are a number of computers which a school is planning to network.

Discuss the advantages and disadvantages, to the students, of having a computer network.

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CENTRE NUMBER

CANDIDATE NUMBER

5846348541

INFORMATION TECHNOLOGY **9626/12**

Paper 1 Theory **October/November 2020**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 90.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Blank pages are indicated.

DC (LK/CGW) 187381/3
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[Turn over

1 **(a)** Tick the most accurate statement regarding the use of vishing.

	✓
Vishing involves mobile (cell) phone text messages persuading people to reveal their personal data	
With vishing, a text message may include a telephone number that connects to an automated voice response system	
Vishing involves installing a piece of malicious software on a customer's computer	
Vishing involves a phone call to persuade people to reveal their personal data	

[1]

(b) Describe, in detail, **three** measures you could take to help prevent you being a victim of vishing.

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..... [3]

(c) Tick the most accurate statement regarding the use of smishing.

	✓
With smishing, a text message is received which may include a website URL and invite the receiver to go to the site	
Smishing is the use of video clips persuading people to reveal their personal data	
Smishing involves using emails to persuade people to reveal their personal data	
Smishing involves receiving a phone call	

[1]

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(d) Describe, in detail, **three** measures you could take to help prevent you being a victim of smishing.

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..... [3]

2 In both parts **(a)** and **(b)** tick the most accurate statement referring to the digital divide.

(a)

	✓
It is often more expensive to purchase goods online than from a shop	
The digital divide only relates to the use of personal computers	
The quality of mobile (cell) phone reception is the same everywhere	
The digital divide can exist between the educated and the uneducated	

[1]

(b)

	✓
The divide between regions of the world is referred to as the regional digital divide	
The digital divide can exist between rich and poor people	
The digital divide does not usually exist between developed and developing countries	
Both urban areas and rural areas tend to have the necessary infrastructure for broadband	

[1]

3 An IT specialist is employed by a car showroom to manage the spreadsheets which record all details of car sales. Here is a spreadsheet showing some of the cars for sale in a showroom.

E18 ⋮ ✗ ✓ *fx* =SUMIFS(E5:E15, A5:A15,I6,G5:G15,">"&I8)

	A	B	C	D	E	F	G	H	I
1	Make	Model	Engine (litres)	Licence	Price	Economy (L/Km)	Year made		
2									
3									
4									
5	Daui	2000	1.8	JPY 648	$25,000	18.0	2016		
6	Tiaf	Visette	1.4	FGB 721	$15,500	16.7	2010		Daui
7	Arajug	EX	1.4	FFA 419	$18,000	13.7	2009		
8	Daui	1000	2	BFK 297	$10,500	12.0	2003		2010
9	Tiaf	Firetips	1.2	GFL 364	$11,000	12.1	2012		
10	Tiaf	Firetips	1.2	HDC 684	$12,500	12.3	2013		
11	Arajug	EX	1.2	JYU 381	$42,500	15.5	2016		
12	Daui	2000	1.8	KJN 384	$28,000	18.5	2017		
13	Tiaf	Delrah	1.6	GXL 347	$19,000	16.3	2013		
14	Arajug	JX	1.2	KNN 478	$47,500	16.6	2017		
15	Tiaf	Visette	1.4	LSA 442	$45,000	19.9	2018		
16									
17									
18					53000				
19									

(a) The IT specialist has entered a formula in E18 after receiving a request from the showroom manager.

(i) Write down the information the showroom manager asked for.

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..... [2]

(ii) Explain, step by step, how the spreadsheet software calculated the result.

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(b) Describe, step by step, how the IT specialist could set up the cell so that the user could just use a mouse to change the input in cell I6.

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..... [3]

(c) The IT specialist accidentally saved the spreadsheet as a csv file.

(i) Explain what a csv file is.

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(ii) Describe the changes the user would see in the spreadsheet after it is opened again using spreadsheet software.

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4 A secretary has been asked to add the data of three new employees to his company's database of employee records.

These include payroll numbers and the date these employees joined the company:

J45786	28/10/2019
K68932	11/11/2019
M40932	30/12/2019

Analyse the effectiveness of limit checks on the upper limits and format checks in validating this data.

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5 A company selling desktop computers uses batch processing to process its customer orders. The company updates the stock master file every week.

Two sets of data are shown below.

The first set represents part of a transaction file containing the computer ID numbers and either the total number of items sold to customers or the amount of new stock delivered to the company in a particular week. The type of transaction is given a code of either S or D. S means that the number of computers in stock needs to be reduced. D means that new stock needs to be added.

The second set represents part of the master file used by the company. This shows the computer ID numbers, the make of computer, the hard disk size and number in stock.

Transaction file

ID_number	Quantity	Type
D20992	40	D
D43487	45	S
D87214	36	S
M47155	43	S
M75860	76	D
R03272	55	S
R62925	92	S
S32249	67	S
S47700	85	D
T49443	61	S

Master File

ID_number	Make	Hard_disk	Number_in_stock
D15609	Dingly	1	162
D20992	Dingly	1.5	145
D43487	Dingly	2	120
D87214	Dingly	3	150
M47155	MBI	1	135
M75860	MBI	2	158
M94382	MBI	4	182
R03272	Rickard	1	147
R55530	Rickard	1.5	126
R62925	Rickard	2	181
S32249	Sensen	2	130
S47700	Sensen	3	167
T49443	Trudel	1	122
T88350	Trudel	4	158

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Using this data, explain the steps involved in updating the master file using this transaction file for both types of transaction. You may assume that the transaction file has already been validated and sorted.

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6 Josefine is in charge of a team of engineers who often work away from the office where she is based. Every week, she keeps in contact with the engineers by means of a web-conference.

Describe how Josefine sets up her web-conference each week.

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7 Describe the roles of a compiler and a linker and explain why both might be necessary in the conversion of a high-level language program.

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8 Select the most appropriate data type for each of the following examples, giving a reason why alternative data types would not be suitable. If the data type is numeric, specify the type of numeric data that would be most appropriate and give a reason. The reasons must be different for each example.

(a) A phone number, such as 028 9649 6437

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..... [1]

(b) A number used to show the fraction of students who like IT, such as 0.125.

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..... [1]

(c) The number of children in a family, such as 4.

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(d) A person's state of employment, such as employed or not employed.

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..... [1]

(e) The amount of money a person earns, such as $100.50.

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..... [1]

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9 Many factors, including the accuracy of collected data, affect the quality of information.

Describe how the method of collecting data may affect the accuracy of the data.

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10 There are several different generic file formats which can be used to save data.

Evaluate, by weighing up the advantages and disadvantages, the use of txt format to save data.

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11 Many companies use management information systems (MIS).

Without describing the role of the MIS manager, describe how other managers within a company would use MIS.

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12 When a person buys a PC it comes with a variety of hardware and software.

(a) Define the term software.

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(b) Explain why the two types of software are needed by a PC user.

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13 Many companies allow workers who work from home to access company data using a virtual private network (VPN).

Discuss the benefits and drawbacks to companies of using a VPN.

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8998142284

INFORMATION TECHNOLOGY **9626/13**

Paper 1 Theory **October/November 2020**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 90.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages. Blank pages are indicated.

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1 **(a)** Tick the most accurate statement regarding input devices.

	✓
A touch screen is easier to use than a keyboard for inputting large amounts of text	
Using a keyboard is the most efficient method of moving a file from one folder to another	
A scanner is an input device used for inputting hard copy documents	
A motor is an input device used for inputting physical variables in a control system	

[1]

(b) Tick the most accurate statement regarding output devices.

	✓
A sensor is an output device used as an actuator in a control system	
A dot matrix printer is the quickest printer for outputting large amounts of text	
A webcam is an output device used to display moving images	
An inkjet printer can be used to output high-quality photographs	

[1]

(c) Hard disk drives, magnetic tape drives and solid-state drives (SSD) are often referred to as secondary storage devices whereas RAM and ROM are often referred to as primary storage devices.

Explain why computers need secondary storage devices.

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2 **(a)** Tick the statement which most accurately refers to the results of using utility software.

	✓
Lossless compression reduces the number of bits by increasing redundant data	
Some information is lost in lossless compression	
Lossy compression reduces the number of bits by identifying unnecessary information and removing it	
Disk defragmentation organises the contents of the disk into the largest number of contiguous regions	

[1]

(b) Tick the statement which most accurately refers to the properties of a hard disk.

	✓
Data access is faster on a hard disk with fragmented files than on a defragmented hard disk	
Disk formatting prepares a data storage device for initial use	
The first stage of formatting is high-level formatting followed by low-level formatting	
Disk partitioning makes the data storage device invisible to an operating system	

[1]

(c) A type of utility software enables files to be deleted.

Explain how the delete utility works and how files can be recovered even after they appear to have been deleted.

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..... [4]

3 A city suffers from the lack of fresh air because of its heavy traffic. A local government agency uses sensors to monitor the level of pollution in the air. The computer system monitors a number of physical variables.

(a) Describe how this system collects and processes this data.

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(b) Compare and contrast the use of monitoring systems with the use of control systems.

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4 Li has just discovered his printer is not responding when he sends a document to print.

Describe how the following components of an expert system are used to diagnose faults with the printer.

(a) User interface

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(b) Inference engine

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(c) Knowledge base

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5 Louise, a teacher, has been asked to write a text book and will need to submit a sample chapter to the publishers. Before she submits this chapter, she wants to proofread it.

Describe how she should visually proofread this chapter.

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6 DVDs and blu-ray discs are optical media which store data and can be used by computers.

Evaluate, by weighing up the advantages and disadvantages, the use of blu-ray discs compared to DVDs for storing computer data.

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7 Define the following types of malware including a description of what each type does.

(a) Rootkit

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..... [3]

(b) Malicious bots

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(c) Ransomware

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8 Describe the differences between a compiler and an interpreter.

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9 Describe, in detail, the characteristics of a WAN and a LAN.

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10 Below are tables from a university database of Metallurgy students. Relationships link the tables.

Students

Studentid	Phone	Second_name
A5384	01217421600	Bhatti
A6528	01213689624	Schmidt
B7238	01214016620	Jones
A8946	01214006451	Gonzales
D9476	01214016620	Jones

Courses

Studentid	Subject	Teacher_code
A5384	Geology of metals	SMI
A6528	Electrochemical Metallurgy	LOU
B7238	Geology of metals	RIC
A6528	Metallography	SAN
D9476	Physical Metallurgy	SAN

Teachers

Teacher_code	Subject 1	Subject 2
LOU	Electrochemical Metallurgy	Metallography
RIC	Geology of metals	Crystallography
SAN	Metallography	Physical Metallurgy
SMI	Electrochemical Metallurgy	Geology of metals
TUR	Crystallography	Geology of metals

(a) Define, exactly, each of the following terms giving examples of the only suitable fields for each, using the tables provided.

(i) Primary key

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(ii) Foreign key

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(iii) Compound key

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(b) Explain, indicating the type of relationship formed, how the relationships were created connecting these tables. You can assume the tables were already populated with data.

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(c) Describe, in detail, the two validation checks that could be carried out on the Studentid field which could **not** be carried out on the Second_name field.

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..... [3]

11 Amir has created a student database query which allows users to type in any name they choose and then the query finds everybody in the database with that name.

Annisa has told him that this is a dynamic parameter query.

Evaluate, by weighing up the advantages and disadvantages, dynamic parameter queries compared with static parameter queries.

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Please turn over for Question 12.

12 There are a number of computers which a school is planning to network.

Discuss the advantages and disadvantages, to the students, of having a computer network.

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..... [8]

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CENTRE NUMBER

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2904424839

INFORMATION TECHNOLOGY **9626/11**

Paper 1 Theory **October/November 2021**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
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- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 90.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages.

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1 Phoebe is collecting data to find the level of pollution in a local river.

(a) Tick the most accurate statement regarding data.

	✓
A group of facts which are used in context is called a set of data.	
Data can only consist of text and numbers.	
Understanding that Paris, London and Berlin are capital cities of countries in Europe makes this a set of data.	
Data cannot be interpreted until it has a context.	

[1]

(b) Tick the most accurate statement regarding the relationship between data and information.

	✓
Data without a context is called information.	
Data is the result of processing information, usually by computer.	
Data that has meaning becomes information.	
Information, when arranged to make meaningful output, becomes data.	

[1]

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Phoebe will use sensors as direct data sources to collect the data. She will place sensors in the river and collect the readings from a computer.

(c) Describe the benefits of gathering data from direct data sources in this scenario.

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(d) Describe the drawbacks of gathering data from direct data sources in this scenario.

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..... [4]

2 **(a)** Tick the most accurate statement referring to the purpose of input devices.

	✓
A sensor is used to store data for later use.	
A magnetic chip reader is used to enter text directly from a bank card.	
A bar code reader is used to read magnetic ink characters.	
A touchpad is used to simulate the behaviour of a mouse when using a laptop computer.	

[1]

(b) Tick the most accurate statement referring to the purpose of output devices.

	✓
A laser printer can only be used to output low volumes of high-quality documents.	
A dot matrix printer can use continuous stationery.	
A plotter is used to output multi-part stationery.	
An inkjet printer is used to output high volumes of low-quality documents.	

[1]

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3 Many online banking applications use the HTTPS protocol, involving encryption, for secure connections.

Evaluate, by weighing up the advantages and disadvantages, the use of HTTPS and the need for encryption in banking websites.

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4 Describe the differences between system software and application software. Include features and examples of each type of software in your answer.

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5 A company owner wishes to buy software to use to produce advertising materials for her company. She can either buy custom written software or off-the-shelf software.

(a) Describe **three** benefits of using custom written software.

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..... [3]

(b) Describe **three** benefits of using off-the-shelf software.

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6 Computers are often used, together with sensors, to control devices in the home.

Evaluate, by weighing up the advantages and disadvantages, the use of control technology in the home.

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..... [6]

7 A company has several IT workers. To help reduce health issues the company encourages workers to take regular breaks.

Describe, in detail, **three** different health issues that can affect these workers. For each issue give one method, other than taking regular breaks, of helping to prevent it. Each method must be different.

Issue 1 .....

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Prevention .....

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Issue 2 .....

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Prevention .....

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Issue 3 .....

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Prevention .....

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..... [6]

8 Many companies have branches in different cities. Branch workers can connect to the main office using the company's virtual private network (VPN) connection.

Describe the characteristics of this type of VPN.

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..... [6]

9 Video-conferencing is one use of the internet.

(a) Describe what is meant by the internet.

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(b) Explain why members of the general public use video-conferencing more now than when it was first introduced.

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10 The spreadsheet below shows the estimated number of visitors to the different countries in the United Kingdom. The figures shown relate to the months of July and August 2019. The formulae used in column D, to add up the number of visitors to each country, are also shown.

	A	B	C	D	E	F	G	H
1		UK Tourism Figures						
2								
3	Country	CCode1		Total number of visitors (millions)		Date	CCode2	Number of visitors (millions)
4	England	En		10.56		July 2019	NI	0.53
5	Northern Ireland	NI		1.14		July 2019	En	5.10
6	Scotland	Sc		1.25		July 2019	Sc	0.62
7	Wales	Wa		0.81		July 2019	Wa	0.43
8						August 2019	En	5.46
9						August 2019	NI	0.61
10						August 2019	Sc	0.63
11						August 2019	Wa	0.38

Spreadsheet

D
UK Tourism Figures
Total number of visitors (millions)
=SUMIF($G$4:$G$11,B4,$H$4:$H$11)
=SUMIF($G$4:$G$11,B5,$H$4:$H$11)
=SUMIF($G$4:$G$11,B6,$H$4:$H$11)
=SUMIF($G$4:$G$11,B7,$H$4:$H$11)

Column D Formulae

(a) Using this spreadsheet as an example, explain, in detail, what is meant by absolute and relative cell referencing and explain why both are used.

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(b) Describe, in detail, how **other** characteristics of spreadsheet software would enable this spreadsheet to be used for modelling purposes. Each characteristic must be different.

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..... [4]

11 Normalisation is often used to refine the structure of a relational database.

Describe **four** advantages of using normalisation.

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..... [4]

12 A computer repair company employs a technician who uses an expert system to diagnose problems with printers. The company also employs a knowledge engineer.

(a) Explain how the technician would interact with the expert system.

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(b) Explain the role of the knowledge engineer.

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..... [3]

13 Here is an extract from a table which is part of a relational database. It requires the use of a compound key.

Studentid	**Second_name**	**Coursecode**
L03127	Gimenez	MAT
L03218	Driscoll	CHE
L03313	De Silva	PHY
L03214	Alicante	BIO
L03127	Gimenez	CHE

Analyse, including its effectiveness, the function of a compound key. Refer to the extract in your answer.

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..... [6]

Please turn over for Question 14.

14 A manager has produced a training video. It has been given to the IT department to edit.

Describe the purpose of the following features of video editing software and how each could be used.

(a) Trimming .....

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(b) Adding captions .....

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6496458222

INFORMATION TECHNOLOGY **9626/12**

Paper 1 Theory **October/November 2021**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
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- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION

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- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Any blank pages are indicated.

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[Turn over

1 A printer driver is a type of device driver.

(a) Tick the most accurate statement regarding printer drivers.

	✓
They are used to control the hard disk drive.	
They handle the translation of requests between the computer and the printer.	
One printer driver can control any printer.	
They ensure that there are no paper jams.	

[1]

(b) Tick the most accurate statement regarding device drivers in general.

	✓
One device driver works with every operating system.	
Device drivers are extra hardware inside the device.	
Upon installation, a device driver detects and identifies the peripheral device.	
A device driver is needed to control the central processing unit of a computer.	

[1]

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(c) Explain the purpose of an operating system.

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..... [6]

2 A user interface is the method by which a user and a computer exchange information and instructions.

(a) Tick the most accurate statement referring to the use of colour in a user interface for users who suffer from colour blindness (colour vision deficiency).

	✓
It is good practice to use complementary colours such as red text on a green background.	
It is good practice to use text in small sizes and weights to emphasise it against a different-coloured background.	
It is good practice to always use as many colours as possible when designing an interface.	
It is good practice to use a soft-coloured background such as grey or cream to allow text of a different colour to be read more easily.	

[1]

(b) Tick the most accurate statement referring to elements of a user interface.

	✓
Radio buttons only allow you to select one option from a list.	
A text box does not allow you to type numbers into it.	
Text should use the same font size whether the information is important or less important.	
It is a good idea to have a different font size for each word in a title.	

[1]

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(c) A developer has been asked to produce software. The software will allow a company's employees to enter data. The user interface will be a graphical user interface (GUI) and must be easy to use.

Describe **four** features of the layout of such a user interface.

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..... [4]

3 **(a)** A company secretary has been given a piece of paper which has the following data written on it.

Johansen, Ingrid, 4578, 05/12/2020, 02493465782

Explain why these are regarded as just items of data. In your explanation give a possible context for the data and describe how this data then becomes information.

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(b) Companies rely on the quality of the information that they use to make decisions.

Describe how each of the following can affect the quality of the information. In your answers, give detail about the possible effects on companies.

(i) The age of the information.

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(ii) The level of detail in the information.

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4 Describe the differences between symmetric and asymmetric encryption, explaining which is more secure.

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5 A company is investigating converting its paper files into computer data files. The company must choose between its existing staff entering the data manually or using an Optical Character Reader (OCR).

Evaluate, by weighing up the advantages and disadvantages, the use of an OCR compared to manual data entry.

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6 Personal data should always be kept confidential and secure. Data can be kept secure by using encryption and by restricting access using methods such as passwords and biometrics.

Describe other methods of keeping data confidential and secure.

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7 There are many types of storage device.

Explain the purpose of storage devices. You should refer to hard disk drives (HDDs), solid-state drives (SSDs) and pen drives in your answer.

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8 The computer configuration in many schools is based on client-server networks.

Discuss the benefits and drawbacks of using a client-server network.

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9 Video-conferencing is one service provided by the internet.

(a) Describe, in detail, **five** other services provided by the internet. Each service must be distinctly different.

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(b) Describe how data is transmitted in a video-conference.

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10 In a medical expert system, the inference engine receives symptoms from the user interface.

Explain how the inference engine uses these inputs to produce possible diagnoses.

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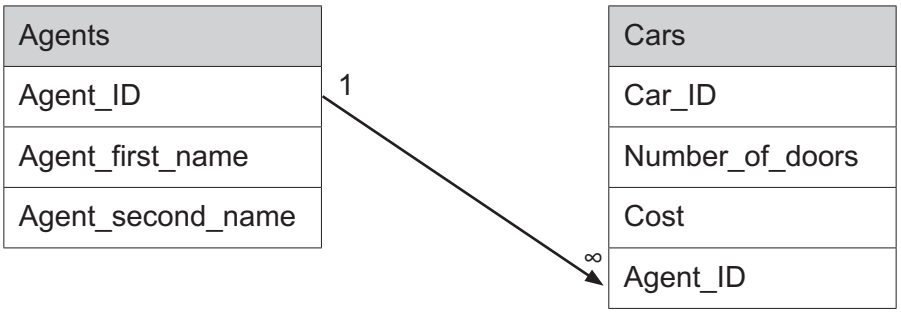
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Please turn over for Question 11.

11 The sales director of a large company buys cars from an agent before issuing them to managers for use as company cars. She needs to keep records of all the agents she deals with and the details of each car. The records are stored as a relational database consisting of two tables.



A typical record from the Cars table would be: S8562T, 5, $39500, B345

(a) Using the information given above, complete this part of the data dictionary for the **Cars** table.

Attribute	Data type	Format	Type of key
Car_ID			

[5]

(b) Evaluate, by weighing up the advantages and disadvantages, the use of a relational database rather than a flat file database in this situation.

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(c) The sales director wishes to find the details of the agent who sold her the car with an ID of R4962Z. She has used a static parameter query to find these details.

Explain why she should have used a dynamic parameter query.

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12 Sound editing software is used to input or record sounds, save them and output the results to various speaker devices.

Describe the use of other features found in sound editing software. Each use must be different.

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8953812122

INFORMATION TECHNOLOGY **9626/13**

Paper 1 Theory **October/November 2021**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 90.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **16** pages. Any blank pages are indicated.

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1 Coding of data, such as G for green and B for blue, has some advantages and disadvantages.

(a) Tick the most accurate statement regarding an advantage of coding data in this way.

	✓
Data can become more difficult to interpret.	
All codes are easy to forget.	
It never leads to the coarsening of data.	
Coding of data means time is saved when entering data.	

[1]

(b) Tick the most accurate statement regarding a disadvantage of coding data in this way.

	✓
Codes cannot be used in a flat file database.	
More mistakes are made when entering data.	
It is possible to run out of meaningful code combinations.	
It is more difficult to use validation rules with the data.	

[1]

(c) A shirt manufacturer uses codes to store information about its shirts. For example, a blue shirt with a neck of 40 cm and a chest of 98 cm could be coded as B4098.

Explain what each part of the code P4395 could mean and why there may be problems with coding shirts in this way. Include in your explanation ways of improving this coding method.

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(d) Other methods are used to convert data. These include encryption and the use of codecs.

Analyse the effectiveness of these two methods.

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2 **(a)** Tick the most accurate statement referring to software.

	✓
An operating system is an example of application software.	
Software consists of data or computer instructions.	
Software consists of RAM and ROM chips.	
A buffer is an example of application software.	

[1]

(b) Tick the most accurate statement referring to system software.

	✓
A spreadsheet is an example of system software.	
A blu-ray disc is an example of system software.	
A printer driver is an example of system software.	
Control software is an example of system software.	

[1]

(c) A company stores all its sales records in a spreadsheet.

Discuss the benefits and drawbacks of using a spreadsheet for this purpose.

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3 Fraudsters sometimes attempt to obtain customers' personal and banking data so they can commit identity theft.

Describe how customers should try to prevent their personal and banking data being obtained by phishing.

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4 Many parents are concerned that their children are not aware of the e-safety issues that arise from using the internet.

Describe the guidance parents should give to make sure their child stays e-safe when using social networking and gaming sites.

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5 Many companies have an intranet for use by their workers.

Describe what a company intranet is.

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6 Many computer hardware companies use expert systems to diagnose computer faults. The inference engine is a key component in an expert system.

Describe, in detail, the other components of an expert system.

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7 All the exams taken in a school are marked out of 80. The school has introduced a system of marking where students are awarded 5 marks for just sitting the exam and then, as normal, are awarded further marks for correct answers.

The marks are stored in a spreadsheet and validated using a range check.

Describe, giving examples, **three** types of test data that could be used to test the range check. For each type of data, describe an improvement that might be necessary as a result of the testing.

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8 After a program is written in a high-level programming language, it has to be converted into a lower level of language.

Evaluate, by weighing up the advantages and disadvantages, the use of an interpreter for this purpose.

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9 Students often confuse the concepts of data, information and knowledge.

Consider the following: 5, 10, 15, 20

Explain, using this example, how data becomes information, and how information can then lead to further knowledge.

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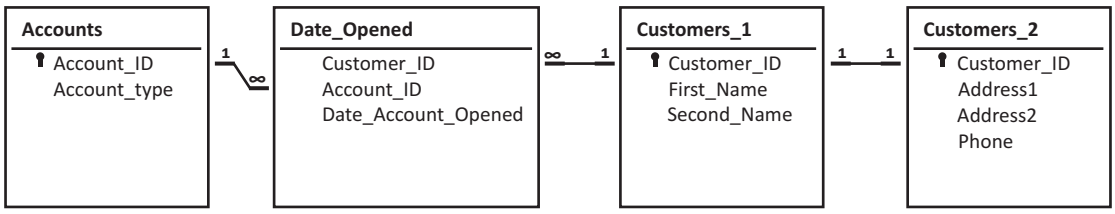
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..... [5]

10 A bank keeps the following information about its customers and accounts in a database:

- all the types of account customers can open
- details of when customer accounts are opened
- the names of each customer
- customers' contact details.

Tables in the database are linked as shown:



(a) Describe, in detail, how the tables are connected and the possible relationships that exist within the database.

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(b) The tables were created with the field names shown. The field names were assigned correctly and then data was entered.

The software rejected the relationships as they were being set up.

Describe **two** reasons the software would have rejected the relationships.

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..... [2]

11 A journalist is researching background information on a story. His editor has provided some hard copy books on the topic. The editor has told the journalist to use the World Wide Web as well, because that could be more reliable than using the books.

Compare and contrast the use of these static and dynamic sources for this research.

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12 Describe what is meant by the following types of database query:

(a) Complex

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(b) Nested

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(c) Summary.

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13 A developer is designing a user interface for one specific user. The developer has been provided with the user's mental model to help with the design.

Describe what a mental model is, and give the benefits to the user of this method of design.

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1624481919

INFORMATION TECHNOLOGY **9626/11**

Paper 1 Theory **October/November 2023**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
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INFORMATION

- The total mark for this paper is 70.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **12** pages. Any blank pages are indicated.

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[Turn over

1 **(a)** Describe the features and the role of a primary key in a database.

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..... [3]

(b) Describe the features and the role of a foreign key in a database.

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2 Two methods of checking the accuracy of data are validation and verification.

Describe the differences between validation and verification.

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3 Companies have the choice of using custom written software or using off-the-shelf software.

Explain the differences between custom written software and off-the-shelf software in terms of:

(a) cost

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..... [2]

(b) level of support

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(c) degree of testing.

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4 Explain how HTTPS websites provide security of data.

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5 Compare static parameter queries with dynamic parameter queries. You must include similarities and differences.

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6 Most control technology systems use actuators and sensors.

(a) Describe the purpose of an actuator.

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(b) Describe how an electromagnetic field sensor works when used in a car park control technology system.

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(c) Describe how an ultrasonic sensor works when a driver is parking a car.

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7 A company employs workers as:

- labourers who are paid $18 per hour
- office workers who are paid $22 per hour
- managers who are paid $30 per hour.

A computer programmer has written part of an algorithm in pseudocode.
The completed algorithm will calculate and output the wage for each worker.

The inputs will be:

- the number of workers
- the job code for each worker (L for labourer, O for office worker or M for manager)
- the number of hours each worker has worked.

If L, O or M is not entered for the job code then the message 'Invalid job code' will be output. The algorithm will terminate when all the workers' details have been entered. It will use the `CASE...ENDCASE` construct to determine the rate of pay.

Complete the algorithm by filling in the missing lines.

```
.....
INPUT number

REPEAT

     INPUT jobcode, hours

.....
          'L' : rate ← 18

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     count ← count + 1

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[8]

8 Explain what is meant by the term ‘data driven’ within the context of expert systems.

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9 Computer models are often used to run simulations when designing a nuclear power plant.

(a) Describe the benefits of using simulations in this scenario.

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(b) Describe the drawbacks of using simulations in this scenario.

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10 A school stores all the records of its students in alphabetical order. Records could be found using sequential access.

(a) Explain what is meant by sequential access.

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..... [1]

The school decides to use the indexed sequential method to search for the records of individual students. An index table contains all the letters of the alphabet in alphabetical order.

(b) Describe how the details of a student called Johnsen would be found.

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[Turn over

11 A company receives orders which are entered into a computer system. The company has decided to use online processing to process these orders so employees enter orders as they are received.

Discuss the advantages and disadvantages of using online processing compared to using batch processing in this scenario.

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12 Analyse the effectiveness of supercomputers.

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5907588310

INFORMATION TECHNOLOGY **9626/12**

Paper 1 Theory **October/November 2023**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
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- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 70.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **12** pages. Any blank pages are indicated.

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[Turn over

1 A company secretary is creating a new database of workers to record:

- their six digit ID number
- the date they joined the company
- their rate of pay per hour up to a maximum rate of $50 per hour.

She has entered data from a source document into the database. There was an error in record 3 and an error in record 4 in the source document. The secretary also made an error when copying record 1.

Source document

Record	ID number	Date joined	Rate of pay
1	147563	13/06/2019	35
2	163459	09/10/2019	28
3	1827697	15/08/2020	31
4	221524	23/12/2020	56
5	256817	27/07/2021	15

Database

ID_number	Date_joined	Rate_of_pay
147563	13/06/2020	35
163459	09/10/2019	28
1827697	15/08/2020	31
221524	23/12/2020	56
256817	27/07/2021	15

Using this data, explain, in detail, the need for verification and validation when entering data into a database.

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2 Mainframe computers are used in banking. Supercomputers are used in climate research.

(a) Explain what is meant by a mainframe computer and by a supercomputer with reference to the types of processing and the processors they each use.

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(b) Explain how supercomputers are used in climate research.

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3 Give brief descriptions of the following types of user interface.

(a) command line interface .....

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(b) gesture based interface .....

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(c) dialogue interface .....

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4 **(a)** Describe how touch sensors are used as part of a control system in nuclear power plants.

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..... [2]

(b) Describe how infrared sensors are used in burglar alarm systems.

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..... [2]

5 Maria wants to use information about the families in her town for a study. She will get the information from an indirect data source, the town's electoral register. This is kept at the local council office. She has decided **not** to use a questionnaire, which is a direct data source, to gather the information she needs.

Describe the suitability of using the electoral register rather than a questionnaire.

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6 There can be a digital divide between people who live in rural areas and people who live in urban areas. The divide can be caused by the lack of fast broadband in rural areas, resulting in limited access to the web.

(a) Explain the effects on people who live in rural areas of having limited access to the web.

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(b) Describe ways to reduce this digital divide.

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7 Louisa, a teacher with limited IT knowledge, has created a simple database of her students. The database contains the names of her students and the results of their latest test.

She prints out the names of the students who have passed the latest test by gaining a mark of 40 or more.

When she created the database, she considered creating a dynamic parameter query to produce her printout. However, she decided to create a static parameter query.

Explain why this was a good decision.

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8 Students often transfer files between different word processing and database packages.

Justify the need for generic file formats for the transfer of files between different packages.

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9 Sukhjinder, a mathematics student, has defined a procedure using pseudocode. It calculates the area of a parallelogram by multiplying the base by the height. All parallelograms will have an area of less than 1000.

The procedure Sukhjinder has written is:

```
PROCEDURE parallelogram(base,height)
area ← base*height
ENDPROCEDURE
```

He has also begun writing an algorithm which calls the procedure and then passes back the area.

The main algorithm accepts inputs of the base and the height of 10 parallelograms, one at a time. It will output the areas of the largest and smallest parallelograms.

Complete the algorithm for Sukhjinder by filling in the missing lines.

```
count ← 0
smallest ← 1000
```

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```
WHILE count < 10
    INPUT base, height
```

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    IF area < smallest
       THEN
```

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```
    ENDIF
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..... [8]

10 Explain why the following techniques could be needed when making videos.

(a) trimming a video clip

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..... [2]

(b) adding subtitles to a video

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..... [2]

(c) adding sound to a video

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11 When data is sent across the internet, it is usually encrypted.

Discuss the benefits and drawbacks of encrypting data which is transmitted over the internet.

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CENTRE NUMBER						CANDIDATE NUMBER				

3454905473

INFORMATION TECHNOLOGY **9626/13**

Paper 1 Theory **October/November 2023**

1 hour 45 minutes

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 70.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **12** pages.

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1 Using a numerical example, explain the differences between data and information.

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..... [4]

2 Irene has produced a document using a word processing package. She has saved it as a text file and also as a separate word processed file.

Give **four** advantages of saving the document as a text file compared with saving the document as a word processed file.

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..... [4]

3 Mainframe computers and supercomputers are specialised types of computer with some similar characteristics.

Explain what is meant by each of the following characteristics.

(a) fault tolerance

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..... [3]

(b) heat maintenance

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..... [2]

Mainframe computers were developed from the need for computers to process data in tasks such as carrying out a census.

(c) Explain why mainframe computers are used to process the data from a census.

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..... [3]

4 A bank stores a database of all its customers. The first field in the database is the account number. The account numbers are different for each customer and consist of 10 digits, for example 7382690643.

New accounts are opened in branches of the bank on a regular daily basis.

(a) Describe **three** errors that a person entering the account numbers of new customers might make. For each one describe an appropriate validation check that would trap the error. A range check or limit check would **not** be suitable. Each validation check must be different.

Error 1 .....

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Validation check 1 .....

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Error 2 .....

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Validation check 2 .....

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Error 3 .....

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Validation check 3 .....

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[6]

When customers open new accounts, the details are gathered together in a transaction file. At the end of the day a branch transmits the transaction file to head office, using the internet.

(b) The transmission of the transaction file is verified using a hash total.

Describe this process, explaining why a hash total is used rather than a control total.

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..... [4]

5 It is essential to calibrate sensors used in any monitoring technology.

Describe what is meant by:

(a) two point calibration .....

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..... [2]

(b) multipoint calibration. .....

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6 When searching a database it is possible to use different types of query.

Describe the differences between simple and complex queries.

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..... [4]

7 Sound editing software has a number of features.

Describe what is meant by:

(a) fade in and fade out

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(b) lossless compression

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(c) lossy compression.

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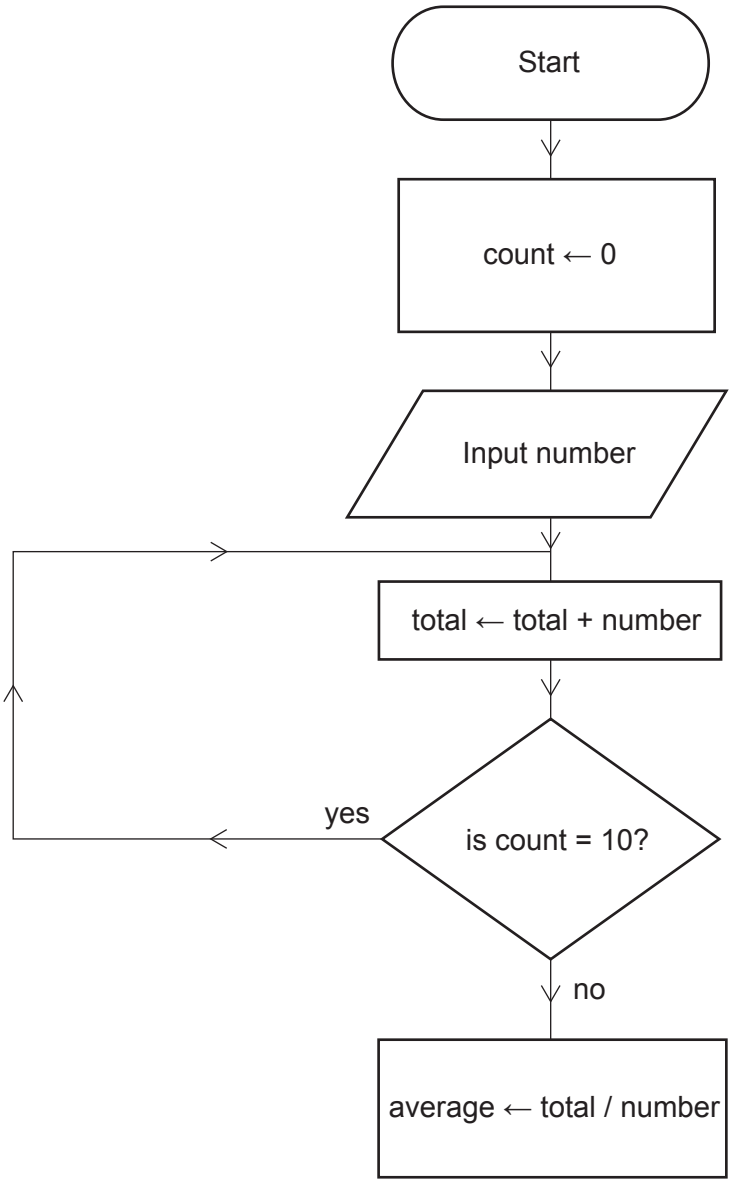
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..... [2]

8 The average (mean) of a set of 10 numbers is calculated by:

- adding the numbers together to get a total
- dividing the total by 10.

Juan has attempted to draw a flowchart of an algorithm which finds the average of 10 numbers. He has made some mistakes and omitted to include some boxes.



Suggest **six** improvements he should make to his flowchart.

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..... [6]

9 A programmer is writing an algorithm in pseudocode to determine whether a number is a factor of another number.

Y is a factor of X if the result of X/Y is a whole number.

In pseudocode `INT(W)` returns the whole number part of W.
If W is the same as `INT(W)` that means W is a whole number.

Complete the algorithm, including the missing first line, in pseudocode. Each `THEN` or `ELSE` statement should be on a separate line to the action resulting from it. The algorithm uses:

- W to store the result of X divided by Y
- Z to store the value of `INT(W)`.

The algorithm must output the values of Y and X together with an appropriate message if Y is a factor of X or if Y is **not** a factor of X.

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W←X/Y

Z=INT(W)
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..... [8]

10 The manager of the payroll department within a company has commissioned a computer programmer to write software to run the payroll.

Justify this decision to use custom written software.

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Please turn over for Question 11.

11 Weather forecasting makes use of computer models.

Analyse the effectiveness of computer models for producing weather forecasts.

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